PHYSICS GRADE 10: ELECTROSTATICS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.4 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity
Sub-Strand	Sub-Strand 3.4: Electrostatics
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Electric charge: types (positive and negative), properties, and the concept of charge conservation – Methods of charging: friction, contact (conduction), and induction – Coulomb's Law: the force between point charges, mathematical expression, and calculations – Electric field: definition, field lines, field patterns around point charges and between parallel plates – Electric potential and potential difference: definitions, units, and relationship to work done – Capacitance: definition, the capacitor, factors affecting capacitance, and the capacitance formula (C = Q/V) – Capacitors in series and parallel: combined capacitance calculations – Energy stored in a capacitor: formula derivation and calculations – Applications of electrostatics: lightning conductors, photocopiers, electrostatic precipitators, spray painting
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the concept of electric charge and the methods by which objects can be charged b) state and apply Coulomb's Law to calculate the electrostatic force between point charges c) describe electric fields and electric potential, and draw field line patterns for various charge configurations d) define capacitance and solve problems involving capacitors in series and parallel circuits e) explain applications of electrostatics in everyday life and evaluate their benefits and risks to society and the environment
Core Competencies	– Communication and Collaboration: Learners discuss and present findings on charge behaviour and field patterns in group investigations, building scientific discourse and peer learning skills – Critical Thinking and Problem Solving: Learners analyse Coulomb's Law, derive capacitance relationships, and solve quantitative problems requiring logical reasoning and mathematical application – Learning to Learn: Learners revise and update their conceptual models of electrostatic phenomena across lessons, developing metacognitive awareness of how their understanding grows – Digital Literacy: Learners use simulations (e.g., PhET Electric Field Hockey) to visualise invisible electric fields, interpreting digital representations of physical phenomena – Self-Efficacy: Learners build confidence by successfully predicting, testing, and explaining electrostatic interactions using both hands-on materials and mathematical tools
Core Values	– Responsibility: Learners consider the responsible use of electrostatic technology (e.g., safe handling of capacitors, grounding

	to prevent static shocks) and its implications for personal and community safety – Respect: Learners respect diverse contributions during group investigations and appreciate the work of scientists — including African researchers — who advanced understanding of electricity – Unity: Learners collaborate in investigative teams, recognising that shared inquiry produces stronger scientific understanding than individual effort alone – Social Justice: Learners evaluate how access to electrostatic applications (e.g., electrostatic precipitators reducing air pollution near Nairobi's industrial areas) can address environmental inequalities in Kenyan communities – Patriotism: Learners connect electrostatics to Kenya's development goals, including safer electrical infrastructure and cleaner industrial processes supporting Vision 2030
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate and refine questions about observed electrostatic phenomena (e.g., why a charged balloon repels another balloon) to drive investigation – Planning and Carrying Out Investigations: Learners design and conduct controlled experiments on charging by friction, contact, and induction using locally available materials (plastic rulers, wool cloth, paper scraps) – Analysing and Interpreting Data: Learners record and interpret data from electroscope deflections and capacitor charge/discharge measurements, identifying patterns and anomalies – Using Mathematics and Computational Thinking: Learners apply Coulomb's Law and capacitance formulae to solve quantitative problems, using proportional reasoning to predict the effect of changing variables – Constructing Explanations: Learners build evidence-based written and diagrammatic explanations of electrostatic behaviour, linking macroscopic observations to microscopic charge models – Obtaining, Evaluating, and Communicating Information: Learners research and present real-world electrostatic applications, evaluating source credibility and communicating findings clearly to peers
Pertinent & Contemporary Issues (PCIs)	– Safety: Learners are taught safe practices around static electricity and high-voltage equipment, linking to personal and community safety in electrical environments – Environmental Education: Learners explore how electrostatic precipitators reduce particulate pollution from factories (e.g., cement plants in Athi River), connecting physics to environmental conservation – Financial Literacy: Learners consider the economic value of electrostatic applications — such as efficient spray painting in manufacturing — and evaluate cost-benefit trade-offs of technology adoption – Life Skills: Learners develop critical thinking and decision-making skills when evaluating risks of static discharge in fuel handling (e.g., petrol stations in Mombasa) and when proposing safety measures
Career Connections	– Electrical and Electronics Engineer: This sub-strand introduces charge, capacitance, and electric fields — foundational concepts for designing circuits, sensors, and communication devices – Atmospheric and Environmental Scientist: Understanding lightning and charge separation in clouds connects directly to meteorological research and weather monitoring at Kenya Meteorological Department stations – Medical Physicist / Biomedical Engineer: Electrostatic principles underpin technologies such as defibrillators, electrocardiograms (ECGs), and electrostatic drug-delivery inhalers used in Kenyan hospitals – Industrial Technician: Knowledge of electrostatic spray painting, powder coating, and precipitators is directly applicable to manufacturing and quality-control roles in Kenyan industries – Telecommunications Engineer: Electric field and capacitance concepts are essential in the design of antennas, signal transmission lines, and network infrastructure supporting Kenya's fibre-optic expansion
Focus for Lessons	This unit investigates electrostatics — the behaviour of stationary electric charges — progressing from the nature of charge itself through forces, fields, and potential, to the practical technology of capacitors. Learners in Kenyan Grade 10 classrooms use locally available materials (plastic combs, wool cloth, aluminium foil) alongside mathematical modelling to build a coherent particle-level

	explanation of phenomena they have experienced but never fully explained. The unit is anchored in a puzzling real-world phenomenon and maintained through a storyline that continuously asks: how do invisible charges produce the forces and effects we observe every day?
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do invisible electric charges build up, exert forces across empty space, and store energy — and how can we harness these abilities to protect communities and power technology in Kenya? KICD KEY INQUIRY QUESTIONS: 1. How do objects acquire electric charge and how does charge behave on different materials? 2. How do charged objects exert forces on one another and on neutral objects at a distance?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Lightning Strike at Ol Kalou Market During a heavy afternoon storm in Ol Kalou, Nyandarua County, a bolt of lightning strikes a metal-roofed market stall, leaving a scorch mark and melting a section of iron sheet — yet the wooden stall next to it is completely untouched. Traders report that just before the strike, the hairs on their arms stood on end and a nearby transistor radio crackled with static. A metal lightning conductor on the district offices 30 metres away directed the discharge safely into the ground with no damage. Students observe a short video reconstruction and a set of photographs showing: (1) the scorch pattern on the iron roof, (2) a leaf electroscope's gold leaves diverging when a charged rod is brought near, and (3) a Van de Graaff generator making hair stand on end. The puzzle is multi-layered — What invisible "thing" built up in the cloud and in the iron roof? Why did the wooden stall survive? How did the lightning conductor "know" where to send the charge? Why did the radio crackle before the strike? Students cannot yet answer these questions; they carry the phenomenon as the engine of inquiry through all nine lessons.
Storyline Thread	Lesson 1 — What Is Happening Before the Lightning Strikes? (Introducing Charge) Students observe the anchoring phenomenon video and photographs of the Ol Kalou lightning event. They record initial predictions on sticky notes for the Driving Question Board (DQB). Through a hands-on friction activity (combs, wool, paper scraps), learners identify two types of charge, the law of charges, and the distinction between conductors and insulators — building the first layer of the particle model. Lesson 2 — How Do Objects Share or Gain Charge Without Rubbing? (Methods of Charging) Learners investigate charging by contact and induction using an electroscope made from aluminium foil and a glass jar. They compare the three methods systematically, construct data tables, and update the DQB: "The iron roof gained charge by induction from the charged cloud — without touching it." The concept of earthing/grounding is introduced using the tanker-chain supporting phenomenon. Lesson 3 — How Strong Is the Push or Pull Between Charges? (Coulomb's Law) Students predict how the force between two charged balloons changes as they are moved apart, then use a torsion-balance demonstration and published data sets to plot F vs. $1/r^2$ graphs. Coulomb's Law is derived from the graph and expressed mathematically. Learners solve problems using Kenyan contexts (e.g., charge separation distance in a storm cloud above the Rift Valley Escarpment). Lesson 4 — How Does a Charge Affect the Space Around It? (Electric Fields and Field Lines) Using iron filings on acetate sheets around charged conductors and a PhET simulation, learners map electric field patterns around

	point charges, dipoles, and parallel plates. They define electric field strength ($E = F/Q$) and connect field line density to force magnitude. The DQB is updated: "The lightning conductor works because it creates a very strong, concentrated field at its pointed tip." Lesson 5 — What Is Electric Potential and Why Does It Matter? (Electric Potential and Potential Difference) Learners use a gravitational analogy (water flowing downhill from the Aberdare Range to the Tana River) to build intuition for potential difference. They define electric potential, potential difference, and the unit (volt), and derive the relationship between work done, charge, and potential difference ($W = QV$). Problem-solving consolidates understanding; the DQB connects potential difference to the "electrical pressure" that drives the lightning bolt. Lesson 6 — What Is a Capacitor and How Does It Store Charge? (Capacitance and the Capacitor) Students construct a simple parallel-plate capacitor from two aluminium foil sheets separated by a plastic bag. They charge it from a 9 V battery, then connect it to an LED to observe discharge. Capacitance is defined ($C = Q/V$), units introduced (farads), and factors affecting capacitance (plate area, separation, dielectric) are investigated systematically. The photocopier supporting phenomenon is revisited. Lesson 7 — How Do Capacitors Behave in Circuits? (Capacitors in Series and Parallel) Learners predict, then verify, how total capacitance changes when capacitors are connected in series versus parallel using a prepared circuit board with known capacitors and a multimeter. They derive the series and parallel formulae through guided mathematical reasoning and solve multi-step problems. The DQB is updated with a circuit diagram showing how capacitor banks store energy in camera flash units and cardiac defibrillators. Lesson 8 — How Much Energy Can a Capacitor Store? (Energy Stored in a Capacitor) Using the charged foil capacitor from Lesson 6 and measured discharge data, learners derive the energy-storage formula ($E = \frac{1}{2}CV^2$). They calculate energy stored in different capacitor configurations and compare this to the energy in a lightning bolt (estimated from current and duration), grounding the abstract formula in the anchoring phenomenon. The precipitator supporting phenomenon is used to calculate the energy input required for industrial air cleaning. Lesson 9 — How Do We Use Electrostatics to Solve Real Problems? (Applications and Unit Synthesis) Student groups research and present one electrostatic application relevant to Kenya (lightning conductors, electrostatic precipitators at Athi River, photocopiers in Kisumu offices, electrostatic spray painting in Nairobi factories, or inkjet printers). Each group updates the class DQB with their application and revises their cumulative particle/field model to show how charge behaviour underpins their chosen technology. A summative written explanation task (using the phenomenon writing rubric) assesses the full unit learning journey.
Supporting Phenomena	– A plastic comb drawn through dry hair in Nairobi's cool highland morning picks up tiny torn pieces of paper — yet the same comb fails to pick up paper after the student washes and wets their hair (supporting charge-by-friction and the role of insulators/conductors) – Petrol tanker trucks on the Nairobi–Mombasa highway trail a metal chain that drags along the tarmac — a safety measure that puzzles many road users who do not understand static discharge risks in fuel handling (supporting grounding and charge dissipation) – Inside a photocopier at a Kisumu office, a drum coated with a photoconductor material selectively attracts black toner powder to reproduce an image — an invisible electrostatic "printing press" operating silently inside a familiar machine (supporting electric field patterns and charge distribution)

	– Smoke and dust particles are removed from the exhaust of the Athi River cement factory by a large metal grid connected to a high-voltage supply — the precipitator cleans air that would otherwise choke nearby Mavoko residents (supporting charge induction, electric fields between plates, and real-world capacitor-like geometry)

LESSON 1 (80 minutes): What Is Happening Before the Lightning Strikes? (Introducing Charge)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon — the lightning strike at Ol Kalou Market — and establish the foundational question: What is electric charge, and how does it build up on objects?
Knowledge	Students will know that: (1) all matter contains positive and negative electric charges; (2) objects become charged when electrons are transferred by friction (triboelectric charging); (3) like charges repel and unlike charges attract; (4) materials are classified as conductors or insulators based on how freely charge moves through them.
Skills	Students will be able to: (1) predict what will happen when charged objects interact; (2) observe and record evidence of charge using an electroscope; (3) use particle-level reasoning to explain charge transfer; (4) pose investigable questions about the phenomenon and record them on a Driving Question Board.
Attitudes	Students will appreciate that: (1) curiosity about natural events such as lightning drives scientific inquiry; (2) careful, evidence-based observation is more reliable than folklore or guesswork; (3) science and technology developed from understanding electrostatics can protect Kenyan communities.
Key Inquiry Question	How do objects acquire electric charge and how does charge behave on different materials?
Purpose in Storyline	This opening lesson anchors the entire unit in the Ol Kalou Market lightning strike. Students cannot yet explain the phenomenon; instead they surface prior ideas, notice what is puzzling, and generate the questions that will drive nine lessons of inquiry. The concept of electric charge — what it is, how it moves, and why some materials resist it — is introduced at the particle level so that every subsequent lesson can build on this foundation.
Safety Notes	Van de Graaff generator (if available): ensure students with pacemakers or hearing aids do not touch the dome. Keep the generator away from flammable materials. When charging rods by friction, ensure wool/fur cloths are dry. No open flames near balloons. Remind students not to touch electrical sockets or live wiring at any point — reinforce that the static charges explored today are very low energy compared with mains electricity or lightning.

B. LESSON OVERVIEW

Lesson 1 launches the nine-lesson Electrostatics unit by immersing students in the anchoring phenomenon: a lightning strike at Ol Kalou Market in Nyandarua County. Students watch a short video reconstruction and examine photographs, then attempt to explain what they see using only prior knowledge. Through a structured Predict–Observe–Explain cycle, students charge objects by friction, interact them with an electroscope and with each other, and build a first particle-level model of electric charge. The lesson closes with the opening of the class Driving Question Board (DQB), where every unanswered puzzle from the phenomenon is posted for the unit ahead. No complete answers are given; the lesson is designed to maximise productive confusion and scientific curiosity.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown a single	Display the Ol Kalou photograph	Strategy: Eliciting Prior Knowledge	Observable indicator: Every

📹 **VIDEO:** [Proof: Field from infinite plate (part 1)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [topical-test-form-3-physics-electrostatics-2-answers] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	dramatic photograph: the scorch mark left on the iron-sheet roof of the Ol Kalou Market stall after the lightning strike. No explanation is given. Working individually for 3 minutes, each student writes answers to three prompt questions in their science notebook: (1) 'What do you think caused this scorch mark?'; (2) 'Why do you think the wooden stall next to it was untouched?'; (3) 'The traders said the hairs on their arms stood on end just before the strike — what do you think caused that?' After individual writing, students share in Think-Pair-Share pairs for 2 minutes, then 5–6 pairs are cold-called to share one idea each with the class. The teacher records all ideas — including incorrect ones — on the 'Our Initial Ideas' section of the whiteboard without evaluating them, labelling it clearly: 'What We Think NOW (Before Investigation).'	on the projector. Say exactly: 'Look carefully at this photograph. This happened during a storm in Ol Kalou, Nyandarua County. I want you to write down what you think caused this — use whatever you already know. You have 3 minutes. No wrong answers at this stage.' [START TIMER — 3 minutes silent individual writing.] After 3 minutes: 'Now turn to your partner and share what you wrote. You have 2 minutes.' [2-minute pair discussion.] Then cold-call at least 6 pairs using the register — do not accept volunteer hands only. Say: 'Amina, what did your pair think caused the scorch?' [WAIT TIME: 10–12 seconds after each name is called before prompting.] Record every response verbatim on the 'Our Initial Ideas' board. When a student says 'electricity' or 'lightning', resist elaborating — say: 'Interesting — let's hold that idea. We'll come back to whether that fully explains it.' After all responses are recorded, say: 'We are going to keep all these ideas visible all lesson. By the end, you will decide which ones your evidence supports.'	Wall. By externalising and displaying all initial ideas without evaluation, the teacher creates a low-stakes record that students can later compare with evidence-based explanations. This activates schema, reveals misconceptions (e.g., 'lightning is fire', 'only metal can burn'), and gives students ownership of the inquiry — they are motivated to find out whose idea the evidence will support.	student has written at least two sentences in their notebook within the 3-minute window (teacher scans the room during pair-share). During cold-call, listen for: (a) use of the word 'electricity' or 'charge' — note which students use particle-level language vs. macroscopic descriptions only. Flag any student who is silent or writes nothing — check in with them during the Observe phase. Target benchmark: at least 70% of students can articulate one plausible cause-and-effect idea about the photograph, even if scientifically incomplete.
Observe Phase 📹 **VIDEO:** [Proof: Field from infinite plate (part 1)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [topical-test-form-3-physics-	Part A — Video (10 min): Students watch the short video reconstruction of the Ol Kalou event (narrated in Swahili/English), which shows: (1) storm clouds building over the Aberdare range; (2) the moment of the strike on the iron roof; (3) a slow-motion replay of a leaf electroscope's gold leaves diverging when a charged ebonite rod is brought near; (4) a Van de Graaff generator causing a student's hair to stand on end at a Nairobi science fair. Students use	Before the video, say: 'As you watch, you are a detective. Your job is to notice everything unusual and to write down every question that pops into your head. Write in your observation guide — at least three observations and two questions. Ready?' [Play video — approx. 8–10 minutes.] After the video, do NOT explain anything. Say: 'Hold your questions — we will use them in a few minutes.'	Strategy: Structured Observation Guide with 'I Notice / I Wonder' Protocol. Separating observation from explanation is a core NGSS Science and Engineering Practice (Planning and Carrying Out Investigations; Analysing and Interpreting Data). The two-column format trains students to defer inference until evidence is gathered, which directly mirrors how scientists like Dr. Richard Leakey approached anomalous field observations — note first,	Observable indicators: (1) Each student's observation guide has at least 3 'I notice' entries and 2 'I wonder' entries after the video — check during station rotations. (2) At Station 2, students can describe the direction of leaf movement ('leaves spread apart / diverge') — if they say 'nothing happened' re-prompt them to look at leaf angle. (3) At Station 3, students should be able to name at least one material that 'blocked' the effect (aluminium foil is the expected

electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	a structured observation guide (printed or on tablet): they must record at least THREE observations in the 'I notice…' column and at least TWO 'I wonder…' questions. Part B — Hands-On Stations (20 min, groups of 4): Each group rotates through three stations over 20 minutes (approx. 6–7 min each, teacher signals rotation). – Station 1 — Balloon & Hair: Students blow up a rubber balloon, rub it vigorously on their hair or a wool sweater, then (a) hold it near small pieces of torn paper, (b) hold it near another balloon rubbed the same way, (c) hold it near an uncharged balloon. They record: 'What happens? Describe exactly what you see.' – Station 2 — Charged Rod & Electroscope: Students rub an ebonite (or PVC pipe) rod with a wool cloth and bring it close to (but not touching) the cap of a leaf electroscope. They observe the gold-leaf divergence. They then try with a glass rod rubbed on silk and record whether the leaves diverge more, less, or equally. – Station 3 — Conductor vs. Insulator Test: Students use a charged balloon to attempt to attract small paper scraps through a sheet of cardboard, a sheet of aluminium foil, a plastic bag, and a wooden board. They record which materials 'blocked' the effect and which did not.	For station rotations: 'You are now moving to the stations. At each station, your job is ONLY to observe and record — do not try to explain yet. I will call rotation every 6 minutes.' [Set visible countdown timer.] Circulate to each station. At Station 1, prompt with: 'Describe exactly what you see — use distance words, direction words.' [WAIT 10 seconds.] At Station 2, ask: 'Does it matter how far away the rod is? Try different distances and record.' At Station 3, ask: 'Is there a pattern in which materials block the effect? What do those materials have in common?' [WAIT 12 seconds before elaborating.] Do NOT confirm or correct observations during this phase — respond to questions with: 'Write that down — that is exactly the kind of question we need for the DQB.'	theorise later. The three-station rotation ensures every student collects multi-modal evidence (kinesthetic, visual, tactile) before constructing explanations.	exception — it may redirect, not block) — this surfaces the conductor/insulator distinction before the Explain phase. Red flag: if a group at Station 1 cannot get the balloon to attract paper, check that the cloth is dry and the rubbing is vigorous — assist quietly without explaining the mechanism.

	All observations are recorded in the 'Observe' column of their notebook table.			
Explain Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Part A — Direct Instruction with Interactive Modelling (12 min): The teacher introduces the two-fluid / two-particle model of charge using a large projected diagram of an atom (labelled in English and Swahili: 'protoni / proton', 'elektroni / electron', 'nyutroni / neutron'). Students are told: (1) All objects contain equal numbers of protons (+) and electrons (−) in their neutral state; (2) electrons can be transferred by friction — the object gaining electrons becomes negatively charged, the one losing electrons becomes positively charged; (3) this is called the triboelectric effect. The teacher uses a physical demonstration: rubbing a PVC pipe on wool cloth, then showing the charged pipe attracting a thin stream of water from a tap (the 'bending water' demo). Students are asked: 'Using the particle model, explain to your partner WHY the water stream bends toward the pipe.' [2 minutes pair discussion, then 3 cold-calls.] Part B — Connecting Back to Ol Kalou (5 min): Students re-read their 'Our Initial Ideas' board. Working individually, they write in their notebooks: 'Using the particle model, explain ONE thing from the Ol Kalou photograph that you could not explain before.' They are told: 'You will not be able to explain everything yet — that is the point. Write what you CAN explain now and put a star next to what still puzzles you.'	Begin Part A with: 'Everything you observed at the stations — the balloon attracting paper, the leaves spreading, the hair standing on end — all of it comes down to one idea: electric charge. Let me show you what is happening at the level of particles you cannot see.' [Project atom diagram.] Walk through the model slowly. After explaining electron transfer, say: 'Now I am going to rub this pipe on this cloth. Predict: which one gains electrons — the pipe or the cloth? Write your prediction in one word. You have 20 seconds.' [WAIT — then reveal answer.] For the bending-water demo: 'Watch carefully. I am not touching the water.' [Perform demo.] Ask: 'Mwangi, what did you observe?' [WAIT 10 seconds.] 'Now, Fatuma, using protons and electrons, why do you think that happened?' [WAIT 12 seconds.] For Part B, say: 'Look back at your initial ideas board. Do not erase anything. Write what you can now explain — and star what still puzzles you.' For Part C, after 15-second wait on the wooden stall question, cold-call three students. If a student says 'wood doesn't conduct', press: 'What does that mean at the particle level? Where are the electrons in wood?' [WAIT 10 seconds.]	Strategy: Bridging Analogy + Return to Phenomenon. The 'free electrons as moving traders in Gikomba Market vs. electrons locked in a wooden crate' analogy grounds the abstract particle model in a Kenyan economic context students recognise. Returning immediately to the Ol Kalou photograph after new instruction (the 'Explain-back-to-phenomenon' move) is a core storyline practice: every new concept must earn its place by explaining something real. Students who can write even a partial particle-level explanation of the phenomenon have demonstrated meaningful conceptual uptake.	Observable indicators: (1) During the bending-water cold-calls, target answer is: 'The negatively charged pipe attracts the positive parts of the water molecules' — accept any answer that references attraction between unlike charges. (2) In Part B written responses, look for use of the words 'electrons', 'transferred', 'positive', 'negative' — highlight 2–3 strong responses to share at the DQB phase. (3) In Part C, the two-sentence conductor/insulator explanation should include a reference to electron movement (or lack of it) — if students write only 'metal conducts, wood does not' without particle-level reasoning, return to the diagram and re-prompt before moving on. Collect notebooks at end of lesson to check all three written responses.

	Part C — Conductor vs. Insulator Explanation (5 min): Teacher draws two diagrams on the board: a metal rod (many free electrons, drawn as moving dots) and a wooden rod (electrons tightly bound, drawn as fixed dots). Students copy and annotate the diagrams. Teacher asks: 'Using these diagrams, why did the wooden stall at Ol Kalou survive while the iron-roofed stall was struck?' [WAIT 15 seconds — cold-call 3 students.] Students write a two-sentence explanation in their notebooks.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Proof: Field from infinite plate (part 1)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Each student is given three sticky notes (or small paper slips). They write one question per slip — questions that the lesson has raised but NOT yet answered — about the Ol Kalou phenomenon. Prompts are projected: 'What still puzzles you about the lightning strike?'; 'What do you still not understand about charge?'; 'What would you need to find out to fully explain what happened at Ol Kalou?' Students have 4 minutes to write their questions individually. Groups of 4 then spend 2 minutes clustering their questions by theme. The teacher facilitates a whole-class gallery where one reporter per group reads out their most important question. The teacher organises questions on the DQB (a large dedicated section of the classroom wall or a projected shared document) under four provisional column headers: (1) 'How Does Charge Build Up?'; (2) 'How Does Charge Move?'; (3) 'Why Did the Conductor Work?'; (4) 'Energy and Forces'. Questions that do not fit are placed under 'Other Puzzles'. The DQB remains	Hand out sticky notes. Say: 'You have 4 minutes. Write one genuine question per slip — questions YOU actually have, not questions you think I want. There are no wrong questions here.' [START TIMER — 4 minutes.] After group clustering: 'Each group, your reporter will read your one most important question aloud.' [Cold-call reporters — do not let only confident students volunteer.] As questions are read, say: 'That is going here under Column 3 — because it is really asking about how charge moves along the conductor. Does that make sense, Chebet?' [WAIT 10 seconds.] After all questions are posted, step back and say: 'Look at this board. Every question on it is real. By Lesson 9, we will have answered every single one of them using evidence — not just my words. Today we answered only this much—' [indicate a small section] '—the rest is what the next eight lessons are for.' Leave the board fully visible. Take a photograph of the board for the ARES platform record.	Strategy: Driving Question Board (DQB) as a Living Epistemic Artefact. The DQB, developed in the NGSS storyline tradition, makes student thinking visible and positions questions — not answers — as the engine of science learning. Organising questions into thematic columns models how scientists categorise unknowns before designing investigations. Students feel authorship over the unit's direction, which research shows increases motivation and metacognitive engagement. Revisiting the DQB at the end of every lesson gives students a tangible sense of growing understanding.	Observable indicators: (1) Every student has written at least two questions on sticky notes — if a student has written a statement rather than a question, prompt: 'Can you turn that into a question that could be investigated?' (2) The quality of questions reveals depth of engagement: surface questions ('What is lightning?') vs. mechanistic questions ('How do charges know to move from the cloud to the roof?') — note which students are already thinking causally. (3) Count how many questions reference the Ol Kalou phenomenon specifically vs. generic electricity questions — target: at least 60% are phenomenon-linked. (4) Photograph or transcribe all DQB questions before students leave — these serve as a unit-long formative baseline.

	visible throughout all nine lessons and is updated at the end of each lesson.			
Model Building Phase 📹 **VIDEO:** [Proof: Field from infinite plate (part 1)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students open to a dedicated 'My Growing Model' page in their science notebooks (or a printed template). They are asked to draw a before-the-strike diagram of the Ol Kalou scene using particle-level representations. Specific instructions are projected: (1) Draw the storm cloud and the iron roof — show where you think the positive and negative charges are located, using + and − symbols; (2) Draw the wooden stall next to it — show where its charges are; (3) Add an arrow or annotation showing why you think the iron roof was struck and the wooden stall was not; (4) Write one sentence: 'I am not yet sure about…' to acknowledge the limits of their current model. Students have 8 minutes to draw individually. Two or three volunteers share their models under the document camera (or on the ARES tablet). The teacher annotates the class version of the model, using student language, and posts it under 'Our Model — Lesson 1' on the classroom wall next to the DQB. Students are told: 'This model will be wrong in some ways — that is fine. We will revise it every lesson until Lesson 9, when it should explain everything on the DQB.'	Say: 'Open to your model page. You have 8 minutes to draw your best current explanation of what was happening at Ol Kalou — at the particle level — just before the lightning struck. Use + and − symbols. Show me where the charges are. This is your model right now — it does not have to be correct.' [START TIMER — 8 minutes. Circulate and observe — do not correct models, but ask: 'Why did you put the − charges there?' and 'What is happening to the electrons in the wooden stall?' WAIT 10–12 seconds after each question.] After 8 minutes, invite 2–3 volunteers: 'Kipchoge, would you share your model under the camera? Talk us through it.' [WAIT 10 seconds.] After each sharing: 'What do others notice that is similar to yours? What is different?' [Cold-call 2 students for comparison comments.] Build the class model on the board. Close by saying: 'This is Lesson 1's model. It is a starting point. By the time we reach Lesson 9, you will be able to explain every arrow, every charge, every material choice — including how the lightning conductor on the district offices 30 metres away sent the charge safely into the ground. That explanation is what the rest of this unit is for.'	Strategy: Progressive Model Revision (NGSS Practice: Developing and Using Models). Committing a model to paper at the start of a unit — before full understanding — has two functions: (1) it surfaces misconceptions in a visible, revisable form rather than leaving them hidden; (2) it gives students a concrete record of how their thinking changes, which builds metacognitive awareness. Using the word 'model' deliberately (not 'diagram' or 'drawing') signals that this is a scientific tool, subject to revision when evidence demands it — exactly as Kenyan meteorologists revise storm-track models when new satellite data arrives.	Observable indicators: (1) Models should show + and − symbols distributed across the cloud and the iron roof — if a student draws only the lightning bolt and no particle representation, they have not yet connected the particle model to the phenomenon; re-prompt during circulation: 'Where did those electrons come from?' (2) The wooden stall should appear in most models — if it is absent, the student has not engaged with the 'why was it untouched?' puzzle; this is a key misconception target for Lesson 2. (3) The 'I am not yet sure about…' sentence reveals the frontier of each student's understanding — collect these as a written formative baseline. Target: at least 80% of models include both + and − symbols and at least one annotation referencing the iron vs. wood contrast. Photograph all models before the next lesson and flag students whose models lack particle-level detail for targeted support in Lesson 2.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 2:

1. PHENOMENON ENGAGEMENT: Did students stay anchored to the Ol Kalou event throughout all five phases, or did the lesson drift into abstract electricity content? If the phenomenon lost its grip during the Explain phase, plan to open Lesson 2 by re-showing the Ol Kalou photograph and asking: 'What can our Lesson 1 model now explain about this image?' 2. DQB QUALITY: Review the photograph of the DQB. Are students asking mechanistic questions ('How do electrons move from cloud to roof?') or surface questions ('What is static electricity?')? If most questions are surface-level, use the first 5 minutes of Lesson 2 to model 'upgrading' one surface question into a mechanistic question as a class. 3. PARTICLE MODEL UPTAKE: When you review the collected notebooks, what proportion of students used + and − symbols in their models AND connected them to the iron/wood contrast? Students who drew the lightning but not the particles need targeted scaffolding at the start of Lesson 2 — consider pairing them with a student whose model showed strong particle-level reasoning. 4. MISCONCEPTIONS SURFACED: Note any recurring misconceptions from the 'Initial Ideas' board (e.g., 'lightning is fire', 'electricity travels through air by itself', 'only wet things conduct'). These should be addressed explicitly — using student evidence from the stations — in Lesson 2's Explain phase. 5. STATION MANAGEMENT: Which station generated the most confusion or off-task behaviour? The electroscope station is often the most challenging — if students could not observe clear leaf divergence, check that rods are dry and that the electroscope leaves are not stuck. Repair or replace equipment before Lesson 2. 6. EQUITY CHECK: Did the same 4–5 students dominate the cold-call responses? Review your register-based cold-call record. In Lesson 2, deliberately start cold-calls from students who were quieter in Lesson 1, and use the Think-Pair-Share structure to give all students processing time before being called upon.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed: (1) a balloon rubbed on hair or wool attracting small paper pieces and repelling another similarly charged balloon; (2) gold leaves of an electroscope diverging when a charged rod was brought near; (3) a charged balloon failing to attract paper through a wooden board but showing an effect through thin cardboard; (4) a video of the Ol Kalou lightning strike reconstruction, a leaf electroscope in action, and a Van de Graaff generator causing hair to stand on end at a Nairobi science fair.
What did I learn?	Students learned that: (1) all matter is made of atoms containing positive protons and negative electrons; (2) objects become electrically charged when electrons are transferred by friction (triboelectric effect) — the object gaining electrons becomes negative, the one losing electrons becomes positive; (3) like charges repel and unlike charges attract; (4) conductors (such as iron sheet) allow electrons to move freely, while insulators (such as wood) do not — this is why the iron-roofed stall at Ol Kalou was struck while the wooden stall was untouched.
How does this explain the phenomenon?	Students began to explain: (1) why the traders' arm hairs stood on end before the strike — charge built up in the cloud induced charge separation in nearby objects, including the hairs, which repelled each other; (2) why the transistor radio crackled — rapidly changing electric fields from charge build-up interfered with the radio signal; (3) why the iron roof was struck — as a conductor, it allowed charge to concentrate and provided a low-resistance path. Students could NOT yet fully explain how charge built up in the cloud, how the lightning conductor directed the strike safely into the ground, or how forces act across empty space — these puzzles are posted on the DQB and will be addressed in Lessons 2–9.

LESSON 2 (80 minutes): Two Types of Charge, the Law of Charges, and Conductors vs. Insulators	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to identify the two types of electric charge, state the law of charges, and distinguish between conductors and insulators using evidence from friction-based investigations.
Knowledge	Learners will know that: (1) there are two types of electric charge — positive and negative; (2) like charges repel and unlike charges attract (law of charges); (3) charge is transferred by friction (triboelectric effect); (4) conductors allow charge to flow freely while insulators resist charge flow; (5) the particle model explains charge in terms of electron transfer.
Skills	Learners will be able to: (1) carry out a friction experiment to charge objects and classify them as positively or negatively charged; (2) use a leaf electroscope to detect charge; (3) construct and annotate a particle-level diagram showing electron transfer during charging by friction; (4) classify common Kenyan materials as conductors or insulator.
Attitudes	Learners will appreciate the value of systematic evidence gathering, show curiosity about invisible phenomena, and demonstrate responsibility when handling laboratory equipment.
Key Inquiry Question	How do objects acquire electric charge and how does charge behave on different materials?
Purpose in Storyline	In Lesson 1 learners anchored the phenomenon and posted initial 'I wonder' questions on the DQB. Lesson 2 provides the first layer of evidence: charge is real, it comes in two types, and the material through which it moves matters — directly explaining why the iron roof at Ol Kalou was struck while the wooden stall survived, and why the traders' arm hairs stood on end.
Safety Notes	No high-voltage equipment is used in this lesson. Ensure the room is reasonably dry — humidity reduces static effects. Caution learners not to rub eyes after handling wool or synthetic fabric. If a Van de Graaff generator is used for demonstration, keep it at least 0.5 m from learners with pacemakers or hearing aids. Dispose of torn paper scraps responsibly.

B. LESSON OVERVIEW
Learners begin by revisiting the Ol Kalou lightning video thumbnail and their Lesson 1 DQB sticky notes. Through a structured 'Predict–Observe–Explain' friction investigation using plastic combs, balloon strips, wool, and nylon fabric, they gather evidence that two distinct types of charge exist and that like charges repel while unlike charges attract. A teacher-led leaf-electroscope demonstration extends the evidence to charge detection. Learners then test an array of Kenyan household materials (iron wire, sisal twine, a piece of iron-sheet offcut, dried maize husk, copper wire from a jua kali workshop, a rubber slipper sole) for conductivity, building a conductor/insulator classification table. The lesson closes with particle-model diagram construction and a DQB update in which learners answer one prior 'I wonder' question and post at least one new question prompted by today's evidence.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Proof: Field from	Learners individually examine two photographs on the ARES screen: (A) the scorch pattern on the iron	Display photographs A and B on the ARES screen. Say: 'Before we do any investigation today, I want	Think-Pair-Share with sentence stems — this strategy externalises prior knowledge and surfaces	Observable indicator: Every learner has a written prediction on their sticky note before the pair-

infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	roof at Ol Kalou Market and (B) the undamaged wooden stall beside it. They read the prompt: 'The iron roof was struck; the wooden stall was not. What is different about these two materials that could explain this?' Each learner writes a silent, individual prediction on a sticky note (or digital equivalent) using the sentence stem: 'I think the iron roof was struck because _____, and the wooden stall was not struck because _____.' Learners then share predictions in a Think-Pair-Share with their shoulder partner before two or three pairs share with the class. Learners keep their predictions visible on their desks throughout the lesson to revisit.	you to think silently on your own. Look carefully at both images. What is different about these two materials — iron and wood — that might explain why one was struck and the other was not? You have 90 seconds of silent thinking time. Write your prediction on your sticky note using the sentence stem on the board.' [Allow 90 seconds of genuine silence — do not circulate or prompt.] After 90 seconds say: 'Now turn to your shoulder partner. You each have 30 seconds to share your prediction. Listen carefully because I will ask some of you to share your partner's idea, not your own.' [Allow 60 seconds of partner talk.] Cold-call 3 pairs: 'Amina, tell me what YOUR PARTNER predicted — not your own idea.' Record key prediction words on the board in two columns: 'Material properties' and 'Something invisible.' Do not confirm or correct. Say: 'Hold on to your predictions. By the end of today you will know whether you were right.'	alternative conceptions (e.g., 'lightning chooses tall things' or 'iron is heavier') without the teacher correcting them, preserving cognitive tension that drives inquiry.	share begins. Teacher scans the room during the 90-second silence. Listen for alternative conceptions during the cold-calls (e.g., 'iron is taller' — note these for explicit address in the Explain phase). Success threshold: At least one learner in each group uses language related to the material itself (not just shape or size), signalling readiness to investigate material properties.
Observe Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools	Groups of 4 carry out a structured three-part friction investigation. PART A — Charging by friction and attraction/repulsion test: Each group has two inflated balloons on strings, one piece of wool cloth, and one piece of nylon fabric (cut from a common Kenyan shopping bag). Learners rub Balloon 1 with wool and Balloon 2 with wool, then slowly bring them together and record what happens. They then rub Balloon 1 with wool and Balloon 2 with nylon, bring them together, and record again. They also bring a charged balloon near a small pile of torn paper scraps ('mapande ya karatasi') and near a	Distribute materials before class begins so no time is lost. Say: 'You are now detectives at the Ol Kalou Market scene. Your job is to gather evidence — not yet explain. Record exactly what you see, hear, or feel. Do not write WHY yet.' [Circulate continuously during Part A. After 8 minutes, pause the class with a hand signal.] Ask the whole class: 'Hands up — which groups saw the two balloons push each other away when both were rubbed with wool?' [Count hands aloud: 'I see 4… 5… 6 groups.'] Then ask: 'Which groups saw them pull toward each other when rubbed with different materials?'	Structured data-recording with a three-column results table — requiring learners to make the Ol Kalou connection in real time (column 3: 'What this means for lightning') prevents the investigation from becoming a disconnected lab activity and keeps the phenomenon as the interpretive frame throughout evidence gathering.	Observable indicators: (1) All groups complete a results table with at least 5 of 6 materials correctly classified as conductor or insulator — teacher checks by circulating and initialling correct tables. (2) During the balloon test, groups can verbally state which combination produced repulsion and which produced attraction before the Explain phase begins. (3) In the column-3 sentences, listen for causal language linking conductors to charge flow and lightning pathways — this reveals readiness for the particle model. Flag any group that classifies sisal or maize husk as a conductor for

2025 🔍 Search ARES for similar readings	thin stream of water from a bottle cap hole. PART B — Comb and paper scraps test: Each learner runs a plastic comb through dry hair five times, then holds it 2 cm above a pile of paper scraps. They record observations and sketch the set-up. PART C — Conductor/insulator classification: Groups receive a set of 6 labelled materials (iron wire, sisal twine, copper wire from a jua kali workshop scrap, dried maize husk, rubber slipper sole offcut, iron-sheet offcut). Using a simple circuit tester (1.5 V cell, bulb, two crocodile-clip leads), they test whether each material allows current to flow (bulb lights = conductor; bulb stays off = insulator) and record in a 3-column table: Material | Conductor or Insulator | What this means for lightning. All observations are recorded in learner notebooks using labelled sketches and a results table.	[Count again.] Say: 'Interesting — so the rubbing material matters. Keep that in your head.' [During Part C, crouch to group level and ask targeted questions:] 'Wanjiku, your bulb is not lighting for the maize husk — what does that tell you about how charge would move through a maize husk in a storm?' [Allow 10-second WAIT TIME before probing further.] 'Now look at your iron-sheet offcut result — how does that connect to the Ol Kalou iron roof?' At the 35-minute mark, call: 'Finish your final table entry. You have 2 minutes to write one sentence under Part C column 3: What this means for lightning.' Cold-call 2 groups to read their sentence aloud before transitioning.		targeted questioning in the Explain phase.
Explain Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	The teacher leads a whole-class sensemaking discussion anchored to the data table. Learners are introduced to the particle model of charge: atoms contain protons (positive, fixed in nucleus) and electrons (negative, free to move in conductors). Charging by friction transfers electrons from one surface to the other — the object gaining electrons becomes negative; the one losing electrons becomes positive. Learners annotate a pre-drawn 'before and after' particle diagram in their notebooks showing a wool cloth rubbing a balloon, with arrows indicating electron movement. They then apply the law of charges — like charges repel, unlike	Begin with data synthesis: display a class results table (compile quickly from 2 group readings). Say: 'Every group found that iron wire and copper wire lit the bulb. Every group found that sisal and maize husk did not. What pattern do you see in the materials that conduct?' [10-second WAIT TIME. Cold-call 3 students who have not yet spoken today.] After learners identify 'metals conduct,' say: 'Let us build a model to explain WHY. Open your notebooks to a fresh page and draw this with me.' [Sketch atom model on board — nucleus with + protons, electron cloud.] 'When I rub this wool on the balloon, electrons — tiny negative particles — jump from the wool	Concept-mapping through annotated particle diagrams — learners physically draw electron transfer arrows, making the abstract model visible and spatially encoded. Connecting every new term back to the Ol Kalou context (iron roof = conductor, wooden stall = insulator) embeds vocabulary in a meaningful narrative rather than isolated definition recall.	Observable indicators: (1) Learner particle diagrams show arrows in the correct direction (electrons move from wool to balloon, not the reverse) — teacher checks 6 random notebooks during circulation. (2) In the Ol Kalou explanation paragraph, all four key terms are used correctly and causally (not just listed). (3) Thumbs up/down response to the 'positive or negative balloon' question — if more than 30% show 'down' or uncertainty, pause and re-draw the model with a different analogy (ugali sharing: if Wanjiru takes ugali pieces from Kamau's plate, who has more? who has less?) before proceeding.

	charges attract — to explain their Part A balloon observations, writing one explanation sentence per observation. The teacher then demonstrates a leaf electroscope: a charged rod brought near causes the gold leaves to diverge; touching the rod to the cap causes even greater divergence; earthing collapses the leaves. Learners sketch and annotate the electroscope in their notebooks. Learners return to their conductor/insulator classification table and add a fourth column: 'Particle-model reason — why does charge move freely / not freely through this material?' Finally, learners revisit the iron-vs-wood comparison and write a full explanation of the Ol Kalou phenomenon in 3–4 sentences, explicitly using the terms: electron transfer, conductor, insulator, law of charges.	onto the balloon. The balloon now has MORE electrons than protons. Is it positive or negative?' [5-second wait — thumbs up/down response.] 'Correct — negative. The wool loses electrons so it becomes…?' [Cold-call: 'Baraka, what does the wool become?'] After electroscope demo: 'Achieng, in one sentence, tell me what the spreading leaves tell you about the charge on the gold foil.' [10-second WAIT TIME.] For the Ol Kalou explanation task, say: 'You have 5 minutes of silent writing. Use these four words — I can see them on the board: electron transfer, conductor, insulator, law of charges. Tell me in 3–4 sentences why the iron roof was struck and the wooden stall was not.' Circulate, read over shoulders, and mark with a tick or a question mark. After 5 minutes, cold-call 2 students to read aloud. Class gives a 'warm and cool' peer response: one strength, one suggestion.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya	Learners return to the class DQB from Lesson 1. Each learner selects one sticky note from the board that today's evidence has helped answer (even partially) and moves it to the 'Evidence is building' column, writing a brief update sentence on the back. They then post at least one new 'I wonder' question generated by today's investigation onto the board. Possible new questions anticipated: 'Can charge build up without rubbing — just by being near something charged?' 'How much charge builds up in a storm cloud?' 'Why does the lightning conductor work — is it just a conductor?' 'Can insulators ever	Point to the DQB and say: 'In Lesson 1 you posted questions you could not yet answer. Look at the board silently for 30 seconds. Find one question — yours or anyone's — that today's evidence has started to answer.' [30-second silence.] 'Pick it up, write one sentence on the back explaining what evidence we gathered today that speaks to that question, and move it to the yellow column labelled Evidence is Building.' [Allow 3 minutes.] Then say: 'Now write a brand-new question — something today's investigation made you MORE curious about — and post it in the blue I Wonder column.' [Allow 2 minutes.] Read	DQB curation as metacognitive anchoring — physically moving a sticky note from 'I wonder' to 'Evidence is building' gives learners a tangible, visible record of conceptual progress. Posting new questions prevents premature closure and sustains the epistemic drive that characterises authentic scientific inquiry.	Observable indicators: (1) Every learner moves at least one sticky note and writes an update sentence — teacher verifies by scanning the DQB after the lesson. (2) New questions posted should show increased sophistication compared to Lesson 1 (e.g., move from 'What is lightning?' to 'Can charge move without touching?') — teacher photographs the DQB board after each lesson for comparison. (3) Listen for questions that reveal persistent misconceptions (e.g., 'Does rubbing create charge from nothing?') — flag these for explicit address in Lesson 3 (conservation of charge).

Curriculum Tools 2025 🔍 Search ARES for similar readings	become conductors?' The class reviews the DQB as a whole, and the teacher annotates which questions will be addressed in upcoming lessons (Lessons 3–9), building a visual 'storyline roadmap.'	4–5 new questions aloud (select ones that preview Lessons 3–5 content). Say: 'Excellent. This question about whether charge can build up WITHOUT rubbing — we call that induction — is exactly what Lesson 4 will investigate. This question about storm clouds connects to Lesson 7. You are building the map of this whole unit.' Draw a simple numbered arc on the board connecting today's DQB questions to lesson numbers 3 through 9.		
Model Building Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners open their 'Cumulative Phenomenon Model' page — a designated double-page spread in their notebooks reserved across all 9 lessons for building an increasingly complete model of the Ol Kalou lightning event. In Lesson 1 this page had only the initial phenomenon sketch. Today, learners ADD three new annotated elements to their model: (1) Label the iron roof and a copper lightning conductor as 'CONDUCTOR — electrons move freely'; (2) Label the wooden stall and sisal rope tying the market awning as 'INSULATOR — electrons do not move freely'; (3) Draw a small inset panel showing the electron transfer during charging by friction (wool + balloon) as an analogy for charge build-up in the storm cloud. Beneath the model, learners write a 'Model Limitation' note: 'My model does not yet explain HOW charge builds up in the cloud or why the charge jumps across the air gap between the cloud and the iron roof.' This limitation statement explicitly seeds the inquiry for Lessons 3 and 4.	Say: 'Open your Cumulative Phenomenon Model page. This page will grow every lesson until Lesson 9, when you should be able to explain every detail of the Ol Kalou event. Today you earned three new pieces of evidence. Add them now.' [Display a model exemplar on the ARES screen — a teacher-drawn version with the three elements labelled but without the limitation note.] 'Copy the structure, but use your OWN words for the labels. You have 6 minutes.' [Circulate and verbally prompt: 'Fatuma, which part of the market scene is a conductor? Add that label with an arrow.' 'Otieno, can you draw the electron-transfer inset — think back to the balloon diagram we drew together.'] After 6 minutes: 'Now write one or two sentences at the bottom under the heading Model Limitation. What can your model NOT yet explain? What question is still open?' Cold-call 2 learners to read their limitation statements. Affirm: 'A good scientific model always knows its own limits. That is exactly what real physicists at the Kenya Meteorological Department do when they model lightning risk	Cumulative annotated modelling — requiring learners to physically revise the same diagram across all 9 lessons makes conceptual growth visible and mirrors the iterative nature of scientific model-building. The 'Model Limitation' sentence is a deliberate NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 2 (Developing and Using Models) move that prevents the illusion of complete understanding and sustains inquiry momentum.	Observable indicators: (1) The model contains all three new elements with correct labels (conductor, insulator, electron transfer inset) — teacher checks 5 models per group during circulation. (2) The Model Limitation statement identifies a genuine gap (charge build-up mechanism or the air-gap discharge) rather than a trivial one ('I did not have enough time') — this reveals whether learners understand the boundary of today's evidence. (3) Exit check: as learners pack up, teacher points randomly to 3 materials on the bench and asks the learner nearest them: 'Conductor or insulator — and how do you know?' A correct, justified answer in under 10 seconds is the mastery indicator for this lesson.

		maps — they note what their model cannot yet predict.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: (1) EVIDENCE OF LEARNING — Did learners' Ol Kalou explanation paragraphs in the Explain phase use all four key terms causally, or did they merely list definitions? What does this tell you about the depth of conceptual change? (2) ALTERNATIVE CONCEPTIONS — Did any learners persist in believing that rubbing 'creates' charge from nothing rather than transferring it? If so, how will you address conservation of charge explicitly at the start of Lesson 3? (3) DQB QUALITY — Compare the sophistication of the new 'I wonder' questions posted today with those from Lesson 1. Are learners asking more specific, mechanistic questions (e.g., about electron movement) or still broad observational ones? What does this suggest about the pacing of Lesson 3? (4) KENYA CONTEXT — Were learners engaged by the jua kali copper wire and iron-sheet offcut materials in the conductor/insulator investigation? Did the local materials strengthen or distract from the conceptual goal? Note any substitutions that worked better. (5) TIMING — The lesson is designed for 80 minutes. Note which phase ran over and adjust the Lesson 3 plan accordingly. If the Model Building phase was rushed, consider beginning Lesson 3 with a 5-minute model gallery walk before introducing new content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that two balloons both rubbed with wool repelled each other, while a balloon rubbed with wool and one rubbed with nylon attracted each other. A charged plastic comb attracted torn paper scraps ('mapande ya karatasi'). Iron wire and copper wire from a jua kali workshop lit a circuit-tester bulb; sisal twine, dried maize husk, and a rubber slipper sole did not.
What did I learn?	There are two types of electric charge — positive and negative. Charging by friction transfers electrons: the object gaining electrons becomes negative; the object losing electrons becomes positive. Like charges repel; unlike charges attract (law of charges). Conductors (e.g., iron, copper) allow electrons to move freely; insulators (e.g., wood, sisal, rubber) resist electron flow. This explains why the iron roof at Ol Kalou was a pathway for the lightning discharge while the wooden stall was not.
How does this explain the phenomenon?	The iron roof at Ol Kalou is a conductor — its free electrons allowed the enormous charge built up in the storm cloud to flow rapidly through the roof, causing the scorch and melting the iron sheet. The wooden stall is an insulator — electrons cannot move freely through wood, so the charge had no pathway through it and the stall was undamaged. The traders' arm hairs stood on end because the large charge nearby exerted an attractive force on the opposite charges in their hair, pulling the hairs toward the cloud — a force-at-a-distance effect that Lesson 3 will explore further.

LESSON 3 (40 minutes): How Strong Is the Push or Pull Between Charges? (Coulomb's Law)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will understand that the electrostatic force between two charges depends on the magnitudes of the charges and the square of the distance between them, enabling them to quantify the invisible forces that drove the Ol Kalou lightning strike.
Knowledge	Learners will state Coulomb's Law (F = kQ1Q2/r^2), identify the variables that affect electrostatic force, and explain what the inverse-square relationship means physically.
Skills	Learners will plot F versus 1/r^2 graphs from a published data set, extract the gradient to determine the force constant k, and solve multi-step Coulomb's Law problems set in Kenyan contexts.
Attitudes	Learners will appreciate that precise mathematical laws govern phenomena that appear sudden and unpredictable, building confidence that science can explain and ultimately protect communities.
Key Inquiry Question	How do charged objects exert forces on one another and on neutral objects at a distance?
Purpose in Storyline	Lessons 1 and 2 established what charge is and how it accumulates on the iron roof by induction. Lesson 3 now asks: once the cloud and the roof both carry charge, how strong is the pull between them and how does that force change as the storm moves closer? This quantitative understanding prepares learners for Lesson 4 (electric fields) by giving them a force law to attach field-line density to.
Safety Notes	No high-voltage equipment is used. Balloon rubbing involves latex — ask about latex allergies before the lesson and substitute plastic film strips for any affected learner. Ensure the classroom floor is clear before students swing or move charged balloons to avoid collision injuries.

B. LESSON OVERVIEW

Students begin by predicting how the force between two charged balloons changes as they are brought closer together or pushed further apart, recording qualitative and semi-quantitative predictions on mini-whiteboards. The teacher then presents a torsion-balance demonstration video (Coulomb's original apparatus) alongside a printed data set of F and r values measured between two charged spheres. Working in pairs, learners calculate 1/r^2, plot F versus 1/r^2, and confirm a straight-line relationship through the origin. They extract the gradient to derive k and write the full Coulomb equation. Guided problem-solving uses a scenario in which a cumulonimbus cloud 800 m above the Rift Valley Escarpment carries a charge separation of 20 C, anchoring the mathematics to the Ol Kalou phenomenon. The lesson closes with a DQB update and a particle-model sketch showing force arrows between the cloud and the iron roof.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Coulomb's law] Source: Khan	Each learner receives two inflated latex balloons on 30 cm strings. Working with a partner, they rub both balloons on their hair or a	Distribute balloons before the lesson begins to save time. Say: Rub your balloon on the wool for about ten seconds — really charge	Elicit-first strategy: students commit to a visible prediction (whiteboard sketch) before seeing any data. The balloon activity	Teacher scans mini-whiteboards for the shape of the predicted graph. Common misconceptions to note: (a) force drops to zero at a

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	piece of wool cloth, then hold them 5 cm apart and describe what they feel and see. They then slowly increase the separation to 20 cm and 40 cm and record qualitative observations. On a mini-whiteboard each pair writes: (1) a word description of how the force changes with distance, (2) a rough sketch of a Force-Distance graph, and (3) a prediction for what happens to the force if the distance is doubled. Learners also connect back to the phenomenon by writing one sentence: What role might this force have played between the storm cloud and the Ol Kalou iron roof?	it up. Now hold both balloons 5 cm apart at eye level. What do you notice? [WAIT 15 seconds for observation, then] Turn to your partner and describe what you feel in your hand. [WAIT 20 seconds for paired talk.] Now slowly move them apart to arm length. What changes? [WAIT 15 seconds.] On your whiteboard, sketch a Force-Distance graph — just your prediction, no right or wrong at this stage. [WAIT 30 seconds.] Ask two or three pairs to hold up their whiteboards. Use a neutral summarising move: I can see a range of ideas here — some of you predict the force drops quickly, others more gradually. Keep that prediction in mind; the data will tell us who is closer.	creates a concrete, embodied sense of the force so that the subsequent graph is grounded in physical experience rather than abstract numbers.	specific distance (linear cut-off thinking), (b) force increases linearly with distance (reversal error), (c) doubling distance halves the force (linear rather than inverse-square intuition). These are not corrected yet — they are flagged privately by the teacher to revisit in the Explain phase.
Observe Phase 📹 **VIDEO:** Coulomb's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects a 3-minute video clip of a torsion-balance demonstration modelled on Coulomb's 1785 experiment, clearly showing how the twist angle (proportional to force) changes as the charged spheres are moved apart. Immediately after the video, each pair receives a printed data table with six rows of experimentally measured values: r (m) values of 0.10, 0.15, 0.20, 0.25, 0.30, and 0.40, with corresponding F (N) values of 8.10e-4, 3.60e-4, 2.03e-4, 1.30e-4, 0.90e-4, and 0.51e-4 (charges Q1 = Q2 = 3.0e-8 C kept constant). Learners add two columns to the table: 1/r^2 (m^-2) and their calculated values. They then plot F (y-axis) versus 1/r^2 (x-axis) on a provided 10x10 cm grid, draw the best-fit straight line, and record the gradient. The teacher circulates, asking probing questions without giving answers.	Before distributing the data sheet, say: You are about to do exactly what Coulomb did in Paris in 1785 — except he had to twist a wire by hand and you have a calculator. Your job is to find the mathematical pattern hiding in this data. [Pause 10 seconds.] When learners have filled in the 1/r^2 column, ask a chosen pair: Why did I ask you to calculate 1/r^2 and not just r? [WAIT 20 seconds for the pair to respond before opening to the class.] Circulate and use the prompt: What does the shape of your graph tell you about how force and distance are related? [WAIT 15 seconds per group.] If a group plots a curved F vs r graph instead, say: That graph is also correct — but is it a straight line? What does that tell you about whether force is directly proportional to r? How could you transform it into a straight line?	Data linearisation as a sense-making tool: by plotting F vs 1/r^2 rather than F vs r, learners generate a straight line through the origin, which is the visual signature of direct proportionality. This bridges the concrete observation (balloons repelling) to the abstract mathematical structure of Coulomb's Law without the teacher simply stating the formula.	Teacher checks three things during circulation: (1) correct calculation of 1/r^2 — a common arithmetic error is computing 1/r and squaring only r in the denominator; (2) correct axis labelling on the graph; (3) the line passes through or very near the origin. Any pair whose line has a large y-intercept is asked: If the charges were zero, what would the force be? Does your line agree with that?

Explain Phase 📹 **VIDEO:** [Coulomb's law](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Newton's Law of Gravity (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	A whole-class discussion synthesises the graph results. The teacher writes the gradient equation on the board: gradient = F / (1/r^2) = F x r^2. Since Q1 and Q2 were held constant in the data, the gradient equals kQ1Q2. Learners calculate k from their own gradient and the given charge values, arriving at approximately 9.0e9 N m^2 C^-2. The teacher then writes the full Coulomb equation: F = kQ1Q2 / r^2, and learners annotate each symbol in their notebooks. Three worked examples follow in increasing complexity: Example 1 — two charges of 1 C separated by 1 m (to confirm k numerically); Example 2 — two charges of 2.0e-6 C and 3.0e-6 C separated by 0.05 m (a bench-scale problem); Example 3 — a Kenyan storm context: a cumulonimbus cloud above the Rift Valley Escarpment carries a positive charge of 20 C at an altitude of 800 m above an iron-roofed market. An equal negative charge has accumulated on the roof below. What is the electrostatic force between them? Learners solve Example 3 independently, then compare with a partner before a class check.	To introduce the formula, say: Your gradient has units of N m^2. We said the gradient equals kQ1Q2. We know Q1 and Q2 from the data sheet. What must k equal? Work it out with your partner. [WAIT 30 seconds.] After pairs report values, say: Every pair in the world doing this experiment gets approximately 9 times 10 to the 9 N m squared per C squared. That number has a name — the Coulomb constant, k. Write it in your notebook; you will use it for the rest of this unit. Before presenting Example 3, ask: Cast your minds back to Ol Kalou. The cloud is high above the market. As the cloud drifts lower during the storm, what happens to r — and therefore to F? [WAIT 20 seconds.] Invite a volunteer to narrate the physics of the approaching storm using only Coulomb's Law language. For Example 3, after independent work say: Compare your answer with your partner. If you disagree, do not just show each other the answer — talk through each step until you agree on where the difference arose. [WAIT 45 seconds for paired reconciliation.]	The teacher uses a phenomenon-return narrative: the mathematics of Coulomb's Law is explicitly mapped onto the timeline of the Ol Kalou storm — cloud descends, r decreases, F increases as 1/r^2, until the insulating air can no longer hold back the discharge. This transforms the formula from an abstract statement into a causal story about the phenomenon learners have been carrying since Lesson 1.	Exit-style probe embedded in Example 3: teacher collects written answers and scans for three error types — (1) not squaring r (using r instead of r^2), (2) unit inconsistency (mixing km and m), (3) sign confusion (reporting a negative force magnitude rather than magnitude with a stated direction). These are addressed verbally before the DQB update.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Coulomb's law](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Each learner writes one sticky note (green = new understanding) to add to the class DQB. The prompt is: Using Coulomb's Law, write one sentence that explains something about the Ol Kalou lightning strike that you could NOT explain at the end of Lesson 2. Examples might include: As the storm cloud moved closer to the iron roof, r decreased and F = kQ1Q2/r^2 increased rapidly, eventually overcoming the resistance of air and triggering the	Say: Look at the DQB. In Lesson 1 we said something invisible was building up in the cloud. In Lesson 2 we said the roof gained charge by induction. Today you can say how strong the pull was — put a number on it. Write your green note and add it. [WAIT 60 seconds for writing.] After notes are placed, read two or three aloud and comment: This group has written a number — 4.5 times 10 to the 9 newtons. That is an enormous	The DQB functions as a visible knowledge-accumulation structure. Green notes mark what Coulomb's Law has explained; orange notes mark what it cannot yet explain. This gap-marking is intentional — it creates productive tension that motivates Lesson 4 (field concentration at sharp points).	Teacher reads all green notes before the next lesson. Any note that restates the formula without connecting it to the phenomenon (e.g., just writes F = kQ1Q2/r^2) is flagged for a one-to-one follow-up at the start of Lesson 4: Can you tell me in words what that formula says about the Ol Kalou storm?

Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	strike. Learners also place one orange sticky note (remaining question) — for example: Does the force depend on what material the charged object is made of? Why does the pointed tip of a lightning conductor matter — is that about force too? These orange notes foreshadow Lessons 4 and 5.	force. No wonder the air broke down and we got a lightning bolt. Now read one orange note aloud: This group is asking about the pointed tip. That is exactly where we are heading in Lesson 4. [WAIT 10 seconds.] Keep that question alive.		
Model Building Phase 📹 **VIDEO:** Coulomb's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final five minutes, learners update their cumulative particle model (begun in Lesson 1 and extended in Lesson 2). They draw a cross-section diagram showing: (1) the storm cloud with a cluster of positive charges at its base, (2) the iron roof with induced negative charges on its upper surface, (3) a double-headed arrow labelled r between cloud and roof, and (4) force arrows labelled F = kQ1Q2/r^2 pointing toward each other on both objects. They annotate: What happens to F as the storm moves closer? and write the answer as a mathematical statement (r decreases, 1/r^2 increases, F increases). Learners store this model page in their science portfolios for use in every subsequent lesson.	Say: Open your model booklet to a new page. You are adding the force layer today — your model already shows charge type and charge transfer; now show the force those charges exert on each other across the gap. Draw the cloud, draw the roof, label the distance, and add force arrows. Make it neat enough that a Form Two student who missed today could read your model and understand Coulomb's Law. [WAIT 3 minutes for drawing.] Circulate and ask one or two learners: Which way is the force arrow on the cloud pointing? Why? If F doubled, what changed — Q or r? This is a micro-formative probe; do not correct publicly, just note.	Progressive model building across the nine-lesson sequence ensures that each lesson deposits a new explanatory layer onto a single growing artefact. By Lesson 9 the model will contain charge type, transfer method, force law, field lines, potential, capacitance, and application — a full mechanistic account of the phenomenon.	Teacher circulates and checks that force arrows point toward the opposite charge (attraction), that both objects carry arrows (Newton's third law), and that the label explicitly references r. Models missing the r label receive the prompt: Coulomb's Law says the force depends on something about the gap between cloud and roof — is that shown in your diagram?

D. TEACHER REFLECTION

After the lesson, consider: Did most learners independently produce the F vs 1/r^2 straight line without scaffolding prompts? Were students able to articulate in their own words why the storm becomes more dangerous as the cloud descends — using Coulomb's Law language rather than everyday language? Did the orange DQB notes show genuine conceptual questions about field concentration and pointed conductors, indicating readiness for Lesson 4? If many learners confused r and r^2 in calculations, plan a five-minute warm-up at the start of Lesson 4 with two rapid numerical checks before introducing the electric field concept. Note which learners have not yet connected the mathematical formula to the physical story of Ol Kalou — these learners may need a paired re-reading of their Lesson 1 sticky notes alongside their Lesson 3 particle model to bridge the gap.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that two charged balloons repel each other and that the repulsion weakens noticeably as they are moved further apart. We also analysed data from a torsion-balance experiment and plotted a straight-line graph of F versus $1/r^2$.
What did I learn?	We learned that the electrostatic force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them: $F = kQ_1Q_2/r^2$, where k is approximately 9.0×10^9 N m^2 C^{-2}.
How does this explain the phenomenon?	This explains why the lightning strike at Ol Kalou became inevitable as the storm cloud descended closer to the iron roof: as r decreased, the force increased as $1/r^2$, rapidly overcoming the insulating strength of air and triggering the discharge.

LESSON 4 (80 minutes): Charging by Contact, Induction, and Earthing — How Did the Iron Roof Gain Charge Without Touching the Cloud?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate and distinguish the three methods of charging — friction, contact, and induction — and explain how the iron roof at Ol Kalou Market acquired charge from the storm cloud without direct contact.
Knowledge	Learners will know that objects can be charged by friction (triboelectric effect), by contact (conduction), and by induction (redistribution of charges without touching); that earthing/grounding removes excess charge to the Earth; and that the separation of charge by induction explains how a neutral conductor near a charged object becomes polarised.
Skills	Learners will be able to: construct and use a homemade aluminium-foil electroscope to detect charge; design and carry out a systematic comparison of the three charging methods; record observations in a structured data table; and use particle-level diagrams to explain charge redistribution during induction.
Attitudes	Learners will appreciate the value of systematic scientific investigation and careful observation in explaining everyday phenomena such as lightning; and will show curiosity about invisible physical processes that have visible, sometimes dangerous, consequences.
Key Inquiry Question	How do objects acquire electric charge and how does charge behave on different materials?
Purpose in Storyline	In Lessons 1–3 learners established that like charges repel and unlike charges attract, that charge is quantised and conserved, and that conductors and insulators behave differently. Lesson 4 now uses those foundations to tackle the central puzzle of the Ol Kalou phenomenon: the iron roof was NOT struck by lightning first — it built up charge by induction from the charged cloud overhead, creating the conditions for the strike. By comparing all three charging methods systematically and introducing earthing, learners gain the mechanistic explanation that links cloud charge → roof polarisation → discharge path → why a lightning conductor works. This lesson is the mechanistic turning point of the storyline.
Safety Notes	No high-voltage equipment is used in this lesson. Ensure learners handle glass jars carefully to avoid breakage. If a Van de Graaff generator is available for demonstration, keep learners at least 30 cm away and ensure no learner with a pacemaker or hearing aid is in the front row. Remind learners never to attempt to replicate lightning experiments outdoors.

B. LESSON OVERVIEW

Learners begin by revisiting the Ol Kalou phenomenon and making a focused prediction: did the iron roof touch the cloud? They then carry out a structured practical investigation using homemade aluminium-foil electroscopes and locally available materials (polythene bags, glass rods, woollen cloth, ebonite/PVC pipe) to compare charging by friction, contact, and induction. A supporting phenomenon — a fuel tanker's drag chain earthing charge to the road — grounds (pun intended) the concept of earthing. Learners construct particle-level diagrams, update their Driving Question Board, and revise their cumulative class model of charge behaviour to now include the induction mechanism that explains the iron roof's role in the Ol Kalou strike.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners are shown a single projected image: a storm cloud over Ol Kalou Market with the iron roof beneath it and a gap of several hundred metres of air between them. The caption reads: 'The cloud is negatively charged. The iron roof has not been touched by anything. Predict: what is the charge on the top surface of the iron roof? What about the bottom surface? Draw and annotate your prediction in your notebook using + and − symbols.' Learners work individually for 3 minutes, then share with a shoulder partner for 2 minutes. Selected learners share with the class.	Display the image and read the caption aloud. Say: 'Before we do any investigation today, I want your honest prediction — not the right answer, your best thinking right now. Use what you already know about charges from our last three lessons.' Allow 3 minutes of silent individual thinking (WAIT TIME enforced — teacher circulates without speaking). After 3 minutes, say: 'Now turn and share your annotated diagram with your partner. Explain your reasoning in at least two sentences.' After 2 minutes of partner talk, cold-call 4 learners: one who drew the roof uniformly negative, one who drew it uniformly positive, one who drew it polarised (negative top, positive bottom or vice versa), and one who drew it neutral. For each, ask: 'What made you choose that distribution?' Do NOT confirm or correct — record all four predictions visibly on the board under the heading 'Our Predictions — We Will Test These.' Say: 'Hold onto your prediction. Our investigation today will give you evidence to decide.'	Predict-then-Investigate: Making predictions before evidence gathering activates prior knowledge about charge behaviour and creates cognitive dissonance when results differ from predictions, deepening retention and understanding of induction.	Observable indicator: Learner annotations show + and − symbols placed with some spatial reasoning (even if incorrect), and learners can articulate at least one reason for their prediction. Flag any learner who cannot place any charge symbols — pair them with a stronger partner during the investigation. Note on a clipboard which of the four prediction types are represented in the class — this informs how to sequence the Explain phase discussion.
Observe Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-	In groups of four, learners construct a simple electroscope (glass jar, aluminium foil strip folded in a V-shape suspended from a cork or card lid with a metal paper clip as the stem, and a small aluminium-foil ball on top). They then systematically investigate three charging methods using a structured data table: (A) Charging by friction — rub a PVC pipe/polythene bag with woollen cloth, bring near the electroscope, observe leaf divergence; then touch the charged rod to the ball	Distribute materials kits (one per group of four). Say: 'Your first task is to assemble your electroscope. I will give you 5 minutes. The diagram is on your activity sheet. The key test: when you bring a charged rod near, the foil leaves should visibly spread apart.' Circulate and check each electroscope — gently touch each ball with a charged PVC pipe to verify it works before groups begin formal testing. When all electroscopes are working, say: 'Now work through Parts A, B, and	Systematic Comparative Investigation with Structured Data Table: By testing all three methods with the same detector (the electroscope) and recording identical variables for each, learners can directly compare outcomes and identify the distinguishing features of each method. The tanker chain is a purposeful anomaly — a real-world earthing example — that creates a 'need to explain' bridge into the Explain phase.	Observable indicators: (1) Learner data tables have entries in every cell for all three methods — blank cells indicate incomplete investigation (prompt: 'What would you need to do to fill that cell?'). (2) Learners correctly identify that in Part C (induction) the electroscope retains charge after the rod is removed — this is the key discriminating observation. (3) Learners note whether the charge acquired by induction is opposite to the inducing rod — listen for this in group talk. Circulate and ask

answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	and observe again. (B) Charging by contact — touch the charged rod directly to the electroscope ball; remove; observe if leaves remain diverged; bring a second charged rod near and observe. (C) Charging by induction — bring the charged rod near (not touching) the electroscope ball; while holding the rod near, touch the ball briefly with a finger; remove the finger first, then the rod; observe the leaves. Each group records: (i) whether leaves diverge, (ii) estimated angle of divergence (small <15°, medium 15–45°, large >45°), (iii) what happens when the charged rod is removed, (iv) whether the electroscope retains charge. A supporting observation: the teacher shows a 90-second video clip or photograph of a fuel tanker on Mombasa Road with a metal drag chain touching the tarmac, and asks: 'What do you think that chain is doing?'	C in order. For each part, complete every column of your data table before moving on. Do NOT skip ahead.' After 20 minutes of investigation, call the class to attention. Project the tanker image/video. Ask: 'What do you think the chain touching the road is doing? Discuss with your group for 90 seconds.' Cold-call 3 groups. Say: 'Keep your answer in your notebook — we will return to this in the Explain phase.' WAIT TIME: enforce 10-second silence before accepting answers during whole-class moments.		each group: 'In Part C, after you removed the rod, did the leaves stay open or close? What does that tell you?' Groups that find the leaves close (suggesting they did not follow the earthing step correctly) should repeat Part C with explicit coaching on the finger-touch-then-remove sequence.
Explain Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners return to whole-class discussion. The teacher guides a structured explanation of all three mechanisms using particle-level diagrams drawn live on the board, connecting each to the Ol Kalou phenomenon. Learners annotate their own diagrams in their notebooks simultaneously. The tanker chain is revisited and formally explained as earthing — the chain provides a conducting path to the Earth (an infinite charge reservoir), removing excess charge and neutralising the tanker, preventing spark ignition of fuel vapour. Learners then write a three-sentence mechanistic explanation of how the iron roof at Ol Kalou became charged without touching the cloud, using the	Begin by returning to the four class predictions recorded in Phase 1. Say: 'Let us look at what your investigation found. Which prediction was closest to your evidence?' Cold-call 3 learners who predicted polarisation and ask: 'What did your electroscope tell you that supports or challenges your prediction?' Then draw three side-by-side diagrams on the board — label them 'Friction,' 'Contact,' 'Induction.' For each, draw a conductor with + and − symbols and animate the electron movement step by step, narrating: 'In friction, electrons are physically scraped from one surface to another — remember our polythene and wool from Lesson 2. In contact, the charged object	Annotated Particle-Level Diagram + Anchored Explanation Writing: Drawing simultaneous with narration forces learners to translate verbal explanation into spatial representation, strengthening conceptual encoding. The three-sentence writing task with mandatory vocabulary use checks whether learners can apply — not merely recall — the mechanism to the specific context of the phenomenon.	Observable indicators: (1) Learner diagrams for induction correctly show NO electrons crossing the gap between the rod and the conductor — electrons only move within the conductor. Flag any learner who draws electrons jumping across the gap (misconception: contact required for charge transfer). (2) Three-sentence explanations use all four target words in a causally correct sequence. (3) When cold-called, learners correctly identify that the charge acquired by induction is OPPOSITE in sign to the inducing charge — this is the key conceptual test. Listen specifically for: 'The negative cloud repelled electrons to the far side of the roof, leaving the near side positive,'

	vocabulary: induction, free electrons, redistribution, earthing.	shares its excess electrons with the conductor it touches. In induction — and this is the key one for our iron roof — the nearby charge PUSHES or PULLS the free electrons in the conductor, but NO electrons cross the gap. The redistribution happens entirely inside the iron roof.' Pause. Ask: 'So if the cloud is negatively charged and it is above the iron roof, which end of the roof will have excess negative charge? Which end will be positive?' WAIT TIME: 12 seconds. Cold-call 4 learners. After establishing the correct polarisation (positive charge accumulates on the top of the roof, closest to the negative cloud), say: 'Now — about that tanker chain. What was it doing?' After learner responses, explain earthing formally: 'When you touched the electroscope ball with your finger in Part C, you connected it to the Earth through your body. Electrons flowed out through you to the ground. When you removed your finger — breaking the Earth connection — the remaining charge was trapped. That is exactly what a lightning conductor does: it keeps the building permanently connected to the Earth, so any induced charge drains away harmlessly.' Instruct learners: 'In your notebook, write three sentences explaining how the Ol Kalou iron roof became charged. Use all four of these words: induction, free electrons, redistribution, earthing.' Allow 4 minutes. Cold-call 2 learners to read their sentences aloud. Correct any misconceptions about electrons crossing the air gap.		which demonstrates correct mechanistic reasoning.
Driving	Each learner receives two sticky	Distribute sticky notes (or direct	Claim-Evidence-Question DQB	Observable indicators: (1) Claims

Question Board (DQB) Creation 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	notes (or writes in a designated section of their notebook if sticky notes are unavailable). On the first, they write one claim they can now make with evidence from today's lesson. On the second, they write one new question that today's lesson raised for them. Learners place or record their claims and questions. The teacher reads selected contributions aloud and physically moves or updates cards on the class DQB — a large display board divided into 'What we know,' 'What we think,' and 'What we still wonder.' The class collectively agrees on a new consensus statement to add to the 'What we know' column: 'The iron roof gained charge by induction from the negatively charged cloud — free electrons in the iron redistributed without any contact — and earthing can drain this charge harmlessly.'	learners to the DQB section in their notebooks). Say: 'On your CLAIM note, complete this sentence starter: I can now explain that… On your QUESTION note, write one thing today's lesson made you more curious about — something we have not yet answered.' Allow 3 minutes. Collect or have learners place notes on the board. Read 5 claims and 5 questions aloud. Ask the class: 'Do we all agree this claim is supported by our evidence today? Thumbs up if yes, sideways if unsure.' For any thumbs-sideways, ask: 'What evidence would make you more confident?' Then read the new consensus statement aloud twice and add it to the DQB 'What we know' column. Point to the driving question: 'How do invisible electric charges build up, exert forces across empty space, and store energy?' Say: 'We have now answered part one — building up. In our next lessons we will tackle the force across space — and later, the energy storage. Your questions on the board will guide us.' Highlight 2–3 learner questions that anticipate Lessons 5–6 content (e.g., 'How strong is the force?' or 'What stops the charge from just spreading everywhere?') and say: 'We will come back to these. Hold them.'	Protocol: Requiring learners to distinguish between a claim (something they can now assert with evidence) and a question (something still unresolved) develops scientific epistemic practices — specifically the understanding that science advances through both answers and productive new questions. Making the DQB update a whole-class negotiation builds collective knowledge ownership.	reference specific evidence from today's investigation (e.g., 'the leaves stayed open after I removed the rod, so the charge was trapped by induction'), not just statements copied from the teacher's board. (2) Questions are generative — they point forward to electric force, field strength, or energy storage, suggesting learners are mentally extending the model. (3) The thumbs assessment reveals the proportion of the class that accepts the consensus claim — more than 3 thumbs-sideways signals that re-teaching of the induction mechanism is needed before Lesson 5.
Model Building Phase 📹 **VIDEO:** Proof: Field from infinite plate (part 1) Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners retrieve their cumulative class model of the Ol Kalou phenomenon (begun in Lesson 1 and revised in Lessons 2 and 3). Working in their groups, they add a new layer to their model: a cross-section diagram of the storm cloud and the iron roof below it, showing (a) the charge distribution in the cloud, (b) the induced polarisation	Return the class cumulative model sheets (or direct learners to their model notebooks). Say: 'Your job in the last 12 minutes is to revise your model to show what you now understand about the roof. You must include all four elements listed on the board: (1) charge in the cloud, (2) induced polarisation in the roof with arrows showing	Cumulative Model Revision: Requiring learners to physically annotate and extend the same model across all nine lessons creates a visible, student-owned record of conceptual development. The four-requirement scaffold ensures the model captures the full mechanistic chain — cloud charge → induction → polarisation	Observable indicators: (1) Electron-movement arrows in the model point in the correct direction relative to the cloud charge (negative cloud → electrons in roof repelled downward → positive charge accumulates on top of roof nearest cloud). Incorrect arrows (electrons attracted toward the cloud) reveal a persistent

for similar videos 📖 **PDF:** topical-test-form-3-physics-electrostatics-2-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	in the iron roof with + and − labels and electron-movement arrows, (c) the earthing path through a lightning conductor into the ground, and (d) a label explaining why the wooden stall next to it was unaffected (no free electrons to redistribute — insulator). Groups annotate their diagrams with the consensus claim from the DQB. Each group shares one addition to their model with the class in a 60-second gallery-walk-style share.	electron movement, (3) the earthing/grounding path, (4) why the wooden stall was unaffected.' Write the four requirements on the board and leave them visible. Circulate and narrate what you see: 'I see Group 3 has put the negative charges at the top of the roof — let us think about which way the cloud pushes the electrons.' Prompt with: 'The cloud is negative. What does a negative charge do to the free electrons in the iron below it?' WAIT TIME: 10 seconds before elaborating. After 10 minutes, conduct a 2-minute gallery walk: each group holds up or posts their revised model. Ask the class: 'What do you notice across all the models? Is there anything we disagree about?' Record any disagreements as open questions on the DQB. Close the lesson by saying: 'Today you moved from seeing lightning as a mysterious force from the sky to understanding the invisible charge mechanics that set it up — step by step, in your model. In Lesson 5, we will quantify the force between charges using Coulomb's Law — and you will be able to calculate how strongly that cloud was pulling on the roof.'	→ earthing path → insulator behaviour — rather than isolated facts.	misconception that must be addressed at the start of Lesson 5. (2) The wooden stall is labelled as an insulator with an explanation referencing the absence of free electrons — not just 'wood does not conduct.' (3) The earthing path is drawn as a continuous conducting line from the roof to the ground, with a label indicating electron flow direction. Groups that omit the earthing path or draw it incorrectly should be seated at the front in Lesson 5 for targeted re-engagement.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners correctly distinguish induction from contact in their data tables — specifically, did they record that induction leaves the electroscope charged after the rod is removed, while an uncharged rod brought near after contact charging causes no further deflection? If most groups missed this, the electroscopes may have been too insensitive — consider adjusting the foil strip length or using a more highly charged rod in Lesson 5's opening review. (2) In the three-sentence writing task, how many learners used all four target vocabulary words in a causally correct sequence? If fewer than 60% of learners achieved this, build in a paired vocabulary rehearsal at the start of Lesson 5 before introducing Coulomb's Law. (3) Did any learners persist in drawing electrons crossing the air gap during induction? This is the single most common misconception in this topic. Prepare a targeted analogy for Lesson 5: 'Think of the cloud as a magnet held near iron filings — the filings reorganise, but nothing jumps across.' (4) Were the homemade electroscopes sufficiently sensitive? If not, which local materials produced the best charge — the PVC pipe with wool, or the polythene bag? Record this for the materials kit preparation guide. (5) Did the tanker chain photograph/video successfully activate learner curiosity about earthing, or did it feel disconnected? If disconnected, in future lessons introduce the tanker chain BEFORE the electroscope

investigation as a motivating puzzle rather than a mid-investigation supporting phenomenon.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that bringing a charged PVC pipe near (without touching) the homemade aluminium-foil electroscope caused the foil leaves to diverge; that touching the ball with a finger while the charged rod was held nearby and then removing the finger BEFORE the rod left the leaves permanently diverged even after the rod was removed; and that the tanker drag chain on Mombasa Road provides a conducting path between the tanker body and the ground.
What did I learn?	Learners learned that charging by induction redistributes charge within a conductor without any physical contact between the charging object and the conductor; that earthing (grounding) removes excess charge by providing a conducting path to the Earth; that the sign of charge acquired by induction is opposite to the sign of the inducing charge; and that insulators (like the wooden stall) cannot be charged by induction because they have no free electrons to redistribute.
How does this explain the phenomenon?	Learners explained that the iron roof at Ol Kalou Market became positively charged on its upper surface through induction — the negatively charged storm cloud repelled free electrons in the iron downward, leaving a positive charge on the surface closest to the cloud — without any contact between the cloud and the roof. This charge separation created the conditions for the lightning strike. A lightning conductor prevents damage by keeping the building connected to the Earth, so any induced charge drains away harmlessly rather than building to a discharge.

LESSON 5 (80 minutes): How Strong Is the Push or Pull Between Charges? — Coulomb's Law	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To quantify the electrostatic force between two point charges using Coulomb's Law and relate it to real-world phenomena such as the lightning strike at Ol Kalou Market.
Knowledge	Students will know that the electrostatic force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them (Coulomb's Law: $F = kq_1q_2/r^2$), and that Coulomb's constant $k \approx 9 \times 10^9$ N m² C^{-2}.
Skills	Students will be able to: (1) apply Coulomb's Law to calculate the magnitude and direction of the electrostatic force between two point charges; (2) analyse how changing charge magnitude or separation distance affects the force; (3) construct and interpret force-vs-distance graphs.
Attitudes	Students will appreciate the power of mathematical models in explaining invisible natural phenomena, and develop curiosity about how quantitative physics underpins safety technologies such as lightning conductors.
Key Inquiry Question	How do charged objects exert forces on one another and on neutral objects at a distance?
Purpose in Storyline	In Lessons 1–4, students established that charges exist, can be transferred, and exert forces. This lesson answers the quantitative question: exactly HOW STRONG is that force — and how does distance explain why the lightning conductor 30 m away could still intercept the discharge before the wooden stall was hit?
Safety Notes	No high-voltage equipment is used in this lesson. If charged rods and pith balls are used for the qualitative hook, ensure rods are dry and free of cracks. Students should not attempt to charge objects near water or damp surfaces. Remind students never to shelter under isolated trees or metal-roofed structures during storms — connect explicitly to the Ol Kalou phenomenon.

B. LESSON OVERVIEW

Students begin by predicting how electrostatic force changes as two charged pith balls are moved apart, sketching their intuitive graphs. They then gather evidence through a structured data-collection activity — using a PhET simulation (or teacher-led demonstration with a torsion balance) to measure force at varying charge magnitudes and separations — and construct F-vs-r and F-vs-q graphs. Through guided sensemaking, they derive the proportionality relationships and are introduced to Coulomb's Law. They apply the law to calculate the force between the storm cloud and the iron roof at Ol Kalou, and compare it with the force on the wooden stall, explaining why charge concentration and distance determined which structure was struck. The lesson closes with a revised class Driving Question Board entry and an updated Consensus Model showing quantitative force arrows.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Coulomb's law] Source: Khan	Students are shown two images side by side: (A) two charged pith balls hanging close together, threads spread wide apart; (B) the	1. Display the two pith-ball images on the projector. Say: 'Look carefully — same charges, different separations. I want you to	Elicit–Compare–Commit: Students commit to a personal prediction before seeing data, which creates productive cognitive dissonance	Observable indicator: All students have a sketch graph in their notebooks before partner discussion begins. Teacher checks

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	same pith balls pulled further apart, threads less spread. They are asked: 'If I double the distance between the two charged balls, what happens to the force between them — does it halve, quarter, or do something else entirely? Draw a sketch graph of Force (y-axis) vs Distance (x-axis) showing your prediction.' Students work individually for 3 minutes, then share with a shoulder partner. The teacher then connects: 'Hold that thought — this same question explains why the lightning conductor 30 metres from the market stall at Ol Kalou could pull the discharge away from the wooden stall just 2 metres away.' A second predict prompt is written on the board: 'If the cloud's charge suddenly doubles during the storm, does the force on the iron roof double, more than double, or less than double?'	predict, in a sketch graph, how force changes as we move the balls further apart. You have 3 minutes — work alone first.' SET TIMER. 2. After 3 minutes, say: 'Turn to your shoulder partner. You have 90 seconds — compare your graphs. Do you agree on the shape of the curve?' WAIT and circulate, noting 2–3 distinct graph shapes students drew (linear, curved downward, steep drop). 3. Cold-call 3 students with different graph shapes: 'Amina, walk us through your graph — what shape did you draw and why?' WAIT 12 SECONDS after each student responds before commenting. 4. Record the three competing predictions on the board under the header 'Our Predictions — We'll test these.' Say: 'We will come back and check every one of these against evidence.'	when evidence contradicts intuitive linear thinking. Recording competing predictions publicly gives students ownership of the scientific question.	for the three common misconceptions: (1) linear decrease, (2) force drops to zero at a fixed distance, (3) correct inverse-square curve. Note which students hold which conception — target cold-calls in Phase 3 to these students to track conceptual shift.
Observe Phase 📹 **VIDEO:** Coulomb's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open the PhET 'Coulomb's Law' simulation (offline-cached on ARES tablets) or follow a teacher-led torsion-balance demonstration. They complete a structured data table in two parts: PART A — Fix $q_1 = q_2 = 1\ \mu C$; vary separation r from 1 cm to 10 cm in steps; record force F. PART B — Fix r = 5 cm; vary q_1 from 1 μC to 6 μC while keeping $q_2 = 1\ \mu C$; record force F. Students plot two graphs by hand on grid paper: Graph 1 — F vs r; Graph 2 — F vs q_1. They annotate each graph with the trend they observe. A Kenya-context data card on the desk reads: 'A cumulonimbus cloud over Ol Kalou can accumulate a charge of up to −40 C at its base. The iron roof of the market stall may carry an induced	1. Before releasing students to the simulation, say: 'You are about to collect real force data. Your job is not just to fill a table — your job is to find the mathematical pattern hiding in those numbers. Write that goal at the top of your data table.' 2. Circulate during data collection. Pause at each group and ask: 'What are you noticing about how fast the force drops as you increase r?' WAIT 10 SECONDS for responses. 3. After Graph 1 is plotted, stop the class: 'Hands up — who got a straight line for F vs r?' (expect some hands). 'Who got a curve?' Say: 'Interesting. Let's look at what kind of curve.' Direct students: 'Now plot log F vs log r. What do you notice about the slope?' WAIT 15 SECONDS. Guide students to identify slope ≈	Data-Driven Pattern Recognition with Log-Linearisation: Plotting raw F-vs-r and then log F-vs-log r guides students from 'it curves' to 'the slope is −2', building the mathematical habit of identifying power-law relationships from data — an NGSS Science and Engineering Practice (Analysing and Interpreting Data). This directly mirrors how Charles-Augustin de Coulomb derived his law experimentally in 1785.	Observable indicators: (1) All data tables are complete with units before graphing begins. (2) Students can state in words the trend in each graph ('as r doubles, F decreases by a factor of four'). (3) Log-log graph has a clearly drawn best-fit line with slope calculated. Teacher looks for students who plot F vs r as a straight line — this signals the target misconception for Phase 3 intervention.

	charge of +0.001 C. The cloud base is approximately 2 000 m above ground.' Students are asked to record this data for use in Phase 3.	−2, hinting at an inverse-square relationship. 4. Cold-call 2 students on Graph 2: 'Kipchoge, describe in one sentence the relationship between force and charge.' WAIT 12 SECONDS. 5. Say: 'Before we explain the pattern — look at the Kenya-context data card. That cloud, that iron roof, that distance of 2 000 m. Hold those numbers. You will need them very soon.'		
Explain Phase 📹 **VIDEO:** Coulomb's law Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces Coulomb's Law: $F = kq_1q_2/r^2$, where $k = 9 \times 10^9$ N m² C⁻². Students annotate the formula in their notebooks, labelling each symbol, its unit, and its meaning. They then work through three scaffolded problems in pairs: PROBLEM 1 (Guided) — Two small charged spheres, $q_1 = 3$ μC and $q_2 = 5$ μC, are placed 0.20 m apart in Nairobi. Calculate the force between them. PROBLEM 2 (Structured) — A trader at Ol Kalou market notices that when she rubs her polyester apron on a warm afternoon, a nearby piece of paper is attracted to it. Model the apron as $q_1 = -8$ μC and the paper as $q_2 = +2$ μC, separated by 0.05 m. Find F. PROBLEM 3 (Anchoring Phenomenon Application) — Using the Kenya-context data card from Phase 2 (cloud charge = −40 C, roof induced charge = +0.001 C, separation = 2 000 m), calculate the attractive force between the cloud and the iron roof. Then calculate the force if the separation were 4 000 m (the cloud is higher). What does this tell us about why lightning chooses the closest tall conductor? Students share answers and discuss: 'Why did the	1. Write $F = kq_1q_2/r^2$ on the board. Say: 'This equation is Coulomb's Law. It was derived by a French physicist, but it describes exactly what happened at Ol Kalou on that stormy afternoon. Let us unpack it together.' Ask: 'What does the r^2 in the denominator tell us about what happens to force when distance doubles?' WAIT 12 SECONDS. Cold-call 3 students. 2. Work Problem 1 together, narrating each step aloud: 'I am writing the formula first, then substituting values with units, then evaluating — I never skip units.' 3. Release pairs to Problem 2. Circulate, asking: 'Is the force attractive or repulsive here — how do you know from the signs of the charges?' WAIT 10 SECONDS per group. 4. Before Problem 3, say: 'This is the Ol Kalou problem. The numbers are real-order-of-magnitude estimates. I want you to calculate, then explain in a sentence: does your answer help explain why the iron roof was struck and not the wooden stall?' 5. Cold-call one student to present Problem 3 solution. Ask the class: 'Do you agree with this answer? Thumbs up, thumbs sideways, or thumbs down — and tell me why.' WAIT 15 SECONDS. 6. Say:	Worked Example → Scaffolded Application → Phenomenon Explanation (I Do–We Do–You Do): This gradual release sequence ensures procedural fluency before conceptual application. Returning to the Ol Kalou numbers transforms the formula from abstract algebra into a physical explanation of a real event, fulfilling the NGSS practice of Using Mathematics and Computational Thinking. Asking students to explain the direction of the force (attractive/repulsive) also integrates earlier lessons on charge types.	Observable indicators: (1) Problem 1 answer: $F \approx 3.375$ N — students who get this correct have correct unit conversion (μC → C). (2) Problem 3 reveals whether students understand the inverse-square effect — the force at 4 000 m should be one quarter of that at 2 000 m. (3) Listen for students invoking both charge magnitude AND distance in their explanation of the phenomenon — this demonstrates integrated understanding. Students who explain only distance or only charge are flagged for targeted questioning.

	lightning conductor on the district offices — not the wooden stall 2 m away from the strike — divert the charge? Use Coulomb's Law in your answer.'	'Notice — when distance doubles, force drops to one quarter. The lightning conductor was taller and more conductive, meaning it had a larger induced charge AND was closer to the cloud's charge centre. Coulomb's Law made the conductor the preferred path.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Coulomb's law](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Newton's Law of Gravity (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes. On the ORANGE note, they write one thing they can now explain about the Ol Kalou phenomenon that they could not explain at the start of this lesson (e.g., 'I can now calculate how strong the pull between the cloud and the iron roof was'). On the BLUE note, they write one new question the lesson has raised (e.g., 'If the force drops so fast with distance, how does the cloud ever build up enough force to strike at all?'). Students post their notes on the class Driving Question Board under the columns 'NOW I CAN EXPLAIN…' and 'NEW QUESTIONS'. The teacher reads three blue notes aloud and marks any that connect to upcoming lessons (e.g., electric field, energy stored in charges) with a star, saying: 'That question is already waiting for us in Lesson 6.'	1. Distribute sticky notes. Say: 'Two minutes — orange note: one thing Coulomb's Law lets you explain about Ol Kalou. Blue note: one new question this lesson has opened up for you. Be specific — write an actual calculation or observation, not just a general statement.' SET TIMER for 2 minutes. 2. Invite students to post notes on the DQB. Briefly read 3 orange notes aloud: 'Listen to what your classmates can now explain.' 3. Read 3 blue notes. For each, ask: 'Does anyone have a partial answer to this question already?' WAIT 10 SECONDS. 4. Point to the overarching driving question on the board: 'How do invisible electric charges build up, exert forces across empty space, and store energy?' Say: 'We have now answered the exert forces part — quantitatively. The store energy part is coming. Keep your blue notes — they are your personal inquiry engine.'	Claim–Evidence–Reasoning on the DQB: The two-colour sticky-note protocol structures metacognitive reflection. Orange notes function as evidence-backed claims; blue notes sustain inquiry momentum. Publicly reading both types models that science advances through both answers and new questions — a core NOS (Nature of Science) principle.	Observable indicators: (1) Orange notes reference specific quantities or relationships (e.g., 'force is four times weaker at twice the distance'). Vague notes ('I understand charges better') signal shallow processing — teacher follows up individually. (2) Blue notes should show conceptual curiosity beyond the lesson scope. Notes that simply restate lesson content indicate students have not yet moved to generative thinking.
Model Building Phase 📹 **VIDEO:** [Coulomb's law](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Students return to their Consensus Model diagram of the Ol Kalou phenomenon (begun in Lesson 1 and revised in each subsequent lesson). Today's revision task: (1) Add quantitative force arrows between the storm cloud and the iron roof, labelling the arrow with the calculated force value from Problem 3 and the formula $F = kq_1q_2/r^2$. (2) Add a second, smaller	1. Say: 'Open your Consensus Model. Today we are adding numbers to the story. Your model should now show not just that there is a force, but how strong it is and why it determines what gets struck.' Give 8 minutes for individual model revision. 2. Circulate and prompt: 'If the wooden stall has no free charges to be induced, what does	Cumulative Model Revision with Quantitative Annotation: Adding calculated values to model diagrams bridges qualitative and quantitative understanding — a hallmark of NGSS Developing and Using Models. Requiring students to justify why the wooden stall has a negligible force arrow forces them to apply Coulomb's Law as a reasoning tool, not just a	Observable indicators: (1) Force arrows on the model are correctly directed (cloud-to-roof attraction shown as arrows pointing toward each other). (2) The force value label on the cloud-to-iron-roof arrow matches the calculated answer from Problem 3 (order of magnitude check: should be on the order of 10^{-1} N for the given values). (3) Students' annotation of

📖 **HTML:** [Newton's Law of Gravity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	force arrow between the cloud and the wooden stall (students must estimate or calculate, noting the wooden stall carries no significant induced charge — so $q_2 \approx 0$), showing why the force on the wooden stall is negligible. (3) Annotate the lightning conductor on the district offices with a note explaining: 'Taller → smaller r → larger F by Coulomb's Law → preferred discharge path.' Students compare their revised model with a partner, checking that force arrows are correctly labelled with direction (attractive = toward each other) and relative magnitude. A class Consensus Model is updated on the front display board by the teacher, incorporating student-suggested annotations.	Coulomb's Law predict for the force on it? Write that on your model.' WAIT 10 SECONDS per student. 3. After 8 minutes, say: 'Turn to your partner. Check each other's models for three things: correct formula label on force arrow, direction of arrow (attractive), and annotation on the lightning conductor. You have 3 minutes.' 4. Cold-call one student to come to the front and add their annotation to the class Consensus Model. Ask the class: 'Does everyone agree with this addition? Raise your hand if you want to suggest a change.' WAIT 12 SECONDS. 5. Conclude: 'Our model now has forces with numbers. Next lesson we will ask — what if there is no second charge there yet? How does the first charge affect the space around it before anything else arrives? That is the electric field — and it will complete the picture of how the lightning conductor works even before the discharge begins.'	calculation recipe. The preview of 'electric field' creates a conceptual bridge to Lesson 6.	the lightning conductor explicitly references Coulomb's Law (smaller r → larger F) — this demonstrates transfer of the mathematical relationship to a new context. Students whose annotations are purely descriptive ('conductor is taller') without invoking the formula are identified for re-teaching in Lesson 6's opening.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students' sketch-graph predictions in Phase 1 reveal a predominantly linear intuition? If so, was the log-log linearisation in Phase 2 sufficient to shift this — or do you need to revisit the inverse-square concept at the start of Lesson 6 with a brief visual (e.g., comparing the area of a sphere at radius r vs 2r)? (2) In Problem 3, did students successfully connect the calculated force value back to the phenomenon narrative — or did they treat it as an isolated calculation? If the Ol Kalou context felt disconnected during Phase 3, consider restructuring: present the phenomenon numbers first and derive the need for a formula from them, rather than teaching the formula first. (3) Review the DQB blue notes tonight. Cluster the new questions into themes (field, energy, conductors, safety) and use these to frame the opening hook of Lesson 6. If more than 30% of blue notes ask about 'what is happening in the space between the charges before the strike', you have strong readiness for the electric-field concept. (4) Check: did all students complete a fully labelled Consensus Model revision? Students who left force arrows unlabelled or omitted the lightning conductor annotation may lack procedural fluency with unit conversion (μC to C) — assign a short practice set before Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (via PhET simulation or torsion-balance demonstration) that the electrostatic force between two charged objects

	decreases as separation increases, and increases as charge magnitude increases. They recorded quantitative data and plotted F-vs-r and F-vs-q graphs, identifying a curved (inverse-square) relationship for distance and a linear relationship for charge.
What did I learn?	Students learned Coulomb's Law ($F = kq_1q_2/r^2$, $k \approx 9 \times 10^9$ N m² C⁻²) and applied it to calculate electrostatic forces in Kenyan contexts, including the force between the storm cloud and the iron roof at Ol Kalou Market. They discovered that doubling the distance reduces the force to one quarter, explaining why taller conductors with smaller separation from the cloud are preferred discharge paths.
How does this explain the phenomenon?	Using Coulomb's Law, students explained why the iron-roofed market stall — not the wooden stall beside it — was struck by lightning: the iron roof had a significant induced charge ($q_2 \neq 0$) and was the closest large conductor, maximising $F = kq_1q_2/r^2$. They also explained why the lightning conductor on the district offices 30 m away could intercept the strike — its height minimised r and its conductive tip concentrated induced charge, producing a larger attractive force than either stall.

LESSON 6 (40 minutes): What Is a Capacitor and How Does It Store Charge? (Capacitance and the Capacitor)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct a simple parallel-plate capacitor, observe charge storage and discharge, define capacitance mathematically, and investigate factors that affect capacitance — advancing the class model of how charge builds up and stores energy before a lightning discharge.
Knowledge	Learners will be able to: (1) define capacitance as the ratio of stored charge to potential difference ($C = Q/V$) and state the unit (farad, F); (2) identify the three factors that affect capacitance of a parallel-plate capacitor — plate area, plate separation, and dielectric material; (3) explain how a capacitor stores energy in an electric field between its plates; (4) connect the charge-separation model in a storm cloud to the concept of a natural capacitor.
Skills	Learners will be able to: (1) construct a simple parallel-plate capacitor from aluminium foil and a plastic bag; (2) charge the capacitor from a 9 V battery and observe LED discharge as evidence of stored energy; (3) systematically vary plate area and separation to record qualitative effects on capacitance; (4) use the formula $C = Q/V$ to solve numerical problems set in Kenyan contexts; (5) record and interpret observations in a structured data table.
Attitudes	Learners will: (1) show curiosity and persistence when constructing and testing their own capacitors; (2) appreciate that everyday materials (aluminium foil, plastic bags) can replicate the physics of sophisticated electronic components; (3) value systematic investigation as a way of establishing causal relationships rather than guessing.
Key Inquiry Question	How do charged objects exert forces on one another and on neutral objects at a distance? — extended here to: How does charge accumulate and remain separated, storing energy that can later be released?
Purpose in Storyline	Lessons 1 to 5 established how charge is created, transferred, and how it exerts forces and creates potential differences. Lesson 6 now asks: what physical arrangement allows large amounts of charge to be separated and held — storing energy until released? This directly models the storm cloud above Ol Kalou as a giant natural capacitor, explains why the lightning bolt is so energetic, and introduces the technology thread (photocopiers, defibrillators, camera flashes) that Lessons 7 and 8 will quantify.
Safety Notes	Use only a 9 V battery — not mains electricity. Warn learners not to touch both foil plates simultaneously after charging to avoid a mild but startling shock. Ensure foil edges are smooth to prevent cuts. Keep the plastic bag dry. Dispose of damaged foil safely.

B. LESSON OVERVIEW

In a 40-minute hands-on lesson learners build a parallel-plate capacitor from two aluminium foil rectangles separated by a thin plastic bag, charge it from a 9 V battery, and discharge it through an LED to confirm stored energy is real and retrievable. They then vary plate area and separation in a guided investigation, recording results in a data table. The formal definition $C = Q/V$ and the unit farad are introduced through guided questioning rather than direct instruction. The lesson closes with learners updating the Driving Question Board by labelling the Ol Kalou storm cloud as a natural capacitor and estimating whether the cloud or their foil capacitor stores more charge for the same voltage.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Capacitors and capacitance] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Capacitor Energy (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Teacher displays a photograph of the Ol Kalou scorch mark alongside a photograph of a small ceramic capacitor removed from a broken radio. Teacher asks: Both of these involve stored electric charge — what do you think the cloud and this tiny component have in common? Learners write individual predictions on sticky notes (2 minutes) answering: (a) Where do you think charge is sitting in each case? (b) What do you think determines how much charge can be stored? Pairs share predictions (1 minute), then three pairs report to the class. Teacher records key prediction words on the board: separation, area, material, voltage, distance. Teacher then shows learners the three construction materials — two identical aluminium foil rectangles (20 cm × 15 cm each), a thin plastic bag cut to the same size, and a 9 V battery with crocodile-clip leads — and asks: Without touching the battery yet, predict whether connecting the battery to the foil sandwich will light the LED. Learners record their prediction and reasoning in their exercise books.	Display the two photographs side by side on the board or printed A3 sheets. Say: We have spent five lessons asking how charge moves and pushes. Today we ask a new question — can charge sit still, stored, waiting? Give 2 minutes of silent individual thinking before pair sharing. Do NOT confirm or deny predictions. After pairs report, say: Keep your prediction — we are going to test it in the next phase. When distributing materials say: Handle the foil gently. We will build something that actually works. WAIT TIME: After asking where charge sits in each case, count silently to 8 before accepting responses — this question requires spatial reasoning that needs processing time.	Prediction recording on sticky notes anchors prior knowledge about charge separation from Lessons 1 to 5 and creates cognitive dissonance when learners discover a passive component can store and release energy. Connecting the storm cloud photograph to a ceramic capacitor from a radio activates the anchoring phenomenon and the supporting photocopier phenomenon simultaneously.	Collect sticky note predictions. Look for: (a) learners who already mention separation of charges on two conductors — these can be stretched with: what happens if the conductors are closer together? (b) learners who think the LED will light immediately when the capacitor is connected — use this misconception productively in the Observe phase.
Observe Phase 📹 **VIDEO:** [Capacitors and capacitance] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Capacitor Energy]	Groups of four construct the foil capacitor: lay one foil rectangle flat, place the plastic bag sheet on top, place the second foil rectangle on top of the plastic bag, and press the sandwich firmly together. Connect the 9 V battery positive terminal to the top foil and negative to the bottom foil using crocodile clips for 10 seconds, then disconnect the battery. Now connect the LED across the two	Demonstrate the construction of one capacitor at the front before groups begin — take 90 seconds to do this clearly, naming each layer. Say: The plastic bag is not just packing — it is doing a specific physical job. Keep that in mind as you build. When groups charge and discharge for the first time, pause the class and say: Everyone look at your LED. What is the evidence that something was	The LED discharge provides a visible, memorable anchor — learners have made something that stores and releases energy from household materials. The two-round systematic investigation (varying area, then separation independently) models controlled experimental design without being told this explicitly. Recording in a structured table prepares learners to identify the mathematical	Circulate and check data tables. Key indicators of understanding: (1) learners record that greater plate area gives a brighter or longer LED flash — correct; (2) learners record that greater separation gives a dimmer or shorter flash — correct; (3) learners can articulate in their own words that the plastic bag keeps the charges separated. Common misconception to address: learners

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	foil plates. Learners observe and record: Does the LED light? For how long? What does this tell us about the foil sandwich? Next, the systematic investigation: (Round 1) Keep the plastic bag separator and vary plate overlap area — full overlap (20 cm × 15 cm), half overlap, quarter overlap — charge for 10 seconds each time, discharge through LED, record relative brightness or glow duration in a table with columns: Plate Area, Observed LED Brightness/Duration, Inference about Charge Stored. (Round 2) Keep full plate area but vary the separator thickness — one plastic bag layer, two layers, three layers — record in the same table format. Teacher circulates and asks probing questions at each group. Learners also hold the charged capacitor briefly and try to detect any force or field with a small piece of tissue paper held near the edge of the foil plates.	stored? After groups have completed both rounds of investigation, ask the class to call out patterns they noticed. Write on the board: MORE plate area leads to MORE or LESS charge stored? GREATER separation leads to MORE or LESS charge stored? Do not give answers yet. WAIT TIME: After writing each pattern question on the board, wait 6 seconds and use cold-calling to get responses from learners who have not yet spoken. If a group reports the LED did not glow at all, say: That is important data too — what might have gone wrong in the construction? Guide them to check for foil contact through a hole in the plastic bag (a short circuit).	relationship C = Q/V in the Explain phase. The tissue-paper field detection connects back to Lesson 4 electric field work.	who say the battery is still powering the LED — ask them to point to the battery and show it is disconnected.
Explain Phase 📹 **VIDEO:** Capacitors and capacitance Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Capacitor Energy (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher guides the class to construct the formal definition from their observations. On the board, teacher writes: The more charge (Q) we pushed onto the plates for a given voltage (V), the more the capacitor stored. Learners are asked: What ratio connects Q and V? Learners suggest Q divided by V. Teacher introduces C = Q/V and the unit farad (F), noting that 1 F = 1 coulomb per volt, and that practical capacitors are measured in microfarads (μF) or picofarads (pF) because a full farad is very large. Teacher presents the parallel-plate formula verbally: capacitance increases with larger plate area, decreases with greater separation, and increases when	Begin by writing Q, V, and C on the board as three unknowns and say: You have already discovered the relationships between these through your experiment. I am going to help you write what you already know in mathematical language. This framing positions the formula as discovered, not delivered. When introducing farad, say: One farad is enormous — it would mean storing one coulomb of charge for every volt. The entire foil capacitor you built probably stores about 200 to 500 picofarads. A picofarad is one million millionth of a farad. The capacitors in a Nairobi hospital defibrillator store about 32 microfarads at 5000 V — and that	Deriving C = Q/V from the observation data (greater area, more charge stored at the same voltage) positions learners as sense-makers rather than receivers. The labelled diagram with field arrows consolidates the Lesson 4 electric field concept inside the capacitor geometry. Kenya-specific contexts (M-PESA terminal, Nairobi defibrillator) make the unit real. Connecting defibrillators to healthcare in Kenya (clinics in rural areas of Nyandarua County) begins building the applications thread that Lesson 9 will deepen.	Exit check on practice problems: (i) expected $C = 2.4 \times 10^{-4} / 12 = 2.0 \times 10^{-5}$ F = 20 μF. (ii) expected $Q = CV = 1200 \times 10^{-6} \times 300 = 0.36$ C. Circulate and look for learners who invert the formula (V/Q instead of Q/V) — prompt them to check units: what are the units of Q/V? Coulombs per volt — and that is what a farad is defined as.

	the space between plates is filled with an insulating material called a dielectric (the plastic bag was acting as a dielectric). Learners copy a labelled diagram of a parallel-plate capacitor into their notebooks, labelling: top plate (positive charge), bottom plate (negative charge), dielectric layer, electric field arrows pointing from positive to negative plate between the plates. Worked Example 1 (Kenyan context): A capacitor in a Safaricom M-PESA point-of-sale terminal stores 0.0045 C of charge when connected to a 9 V battery. Calculate its capacitance. Solution: C = Q/V = 0.0045 / 9 = 0.0005 F = 500 μF. Worked Example 2: A student doubles the plate area of their foil capacitor. If the original capacitance was 200 pF, what is the new capacitance? Answer: 400 pF. Learners solve two practice problems in their books: (i) A capacitor stores 2.4×10^{-4} C at 12 V — find C. (ii) A camera flash capacitor has C = 1200 μF charged to 300 V — find the charge stored.	stores enough energy to restart a heart. After worked examples, circulate during the two practice problems. WAIT TIME: After asking what ratio connects Q and V, wait 8 seconds in silence before accepting responses. After giving the answer to Worked Example 1, wait 5 seconds and then ask: Does this answer make sense given what you observed with the LED? Why?		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Capacitors and capacitance Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Capacitor Energy (Flexbook) Source: CK-12 🔍 Search ARES	Each group receives one large sticky note. They are given the prompt: How does what we learned today help us explain the lightning strike at Ol Kalou Market? Groups have 3 minutes to write one sentence that connects capacitance to the phenomenon. Responses are posted on the DQB under the heading Lesson 6 Evidence. Teacher reads two or three aloud. The class then collectively labels the storm cloud diagram (pinned to the DQB since Lesson 1) with new annotations: the base of the cloud is labelled negative plate, the iron roof is	Say: The DQB has been asking since Lesson 1 what invisible thing built up in the cloud and the iron roof. We can now answer that more precisely. Use the word capacitor in your sentence if it fits. After groups post their sticky notes, read the best two or three aloud and ask the class: Do these answers feel complete? What are we still not sure about? Add the new sub-question about energy in a different colour sticky note to signal it is a question generated by this lesson rather than carried from the start. WAIT TIME: After posing the cloud-roof capacitor question,	Labelling the storm cloud diagram with capacitor vocabulary physically integrates the new concept into the running phenomenon model. Posting the energy question as a DQB arrow pointing forward to Lesson 8 gives learners a sense of the unfolding storyline and motivates continued engagement. Using the same diagram annotated across six lessons creates a visible record of growing model sophistication.	Read the sticky note sentences for quality of connection: strong responses will mention charge separation, two conducting surfaces, and dielectric (air or plastic). Weak responses will be vague (the cloud stored electricity). Use weak responses as teaching moments without embarrassing groups — say: That is a start — what word from today can we add to make it more precise?

for similar readings	labelled positive plate (induced), the air gap is labelled dielectric (poor), and the charge stored is labelled Q = CV. Teacher asks: If the air gap between the cloud and the roof is about 500 metres and the potential difference is 100 million volts, and if we treat the cloud-roof system very roughly as a giant capacitor, what does that tell us about the energy waiting to be released? Learners do not calculate this yet — that is Lesson 8 — but they record the question on the DQB as a new sub-question: How much energy is stored in a lightning bolt?	wait 10 seconds — this is a novel application of a brand-new concept and requires significant transfer time.		
Model Building Phase 📹 **VIDEO:** Capacitors and capacitance Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Capacitor Energy (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually update their personal particle/field model diagram (started in Lesson 1 and revised each lesson). Today they must add: (1) a parallel-plate capacitor symbol and label it with C = Q/V; (2) an arrow showing energy stored in the electric field between the plates; (3) a comparison annotation: storm cloud above Ol Kalou = natural capacitor with very large plate area and very large separation — what does that suggest about its capacitance? Learners share their updated models with a partner and explain one new connection they have drawn. Teacher selects two contrasting models to share with the class — one that correctly shows the field inside the capacitor and one that needs refinement — and leads a brief whole-class discussion (2 minutes) on what makes a model more or less useful.	Say: Your model is a scientific tool, not a drawing exercise. Every time you add to it, ask yourself: does this new piece connect to something already on the model? If it floats alone with no arrows connecting it, it is not yet part of the model. When selecting models to share, ask the owner: Can you walk us through one connection you drew today? Do not ask learners to share a model they are uncertain about unless they volunteer. After the partner share, say: In Lesson 8 we are going to put a number on the energy stored in your foil capacitor and compare it to the lightning bolt. Keep your model — we will need it. WAIT TIME: After asking what the large plate area and large separation of the storm cloud suggest about its capacitance, wait 8 seconds — this requires learners to apply the inverse relationship between separation and capacitance, which is a transfer task.	The cumulative model diagram is the primary artefact tracking conceptual growth across the unit. Asking learners to add the capacitor as a node with connections to the electric field (Lesson 4) and potential difference (Lesson 5) reinforces that concepts are not isolated facts but part of an integrated explanatory framework. The contrast between the foil capacitor (tiny C, small energy) and the storm cloud (enormous charge separation, enormous energy) sets up Lesson 8 quantitatively without doing the mathematics yet.	Check that individual model diagrams show: electric field arrows pointing from positive to negative plate inside the capacitor; a label C = Q/V; at least one arrow connecting the capacitor to either the electric field node (Lesson 4) or the potential difference node (Lesson 5). Learners who draw a capacitor in isolation with no connections need a prompt: what does the electric field between the plates remind you of from Lesson 4?

D. TEACHER REFLECTION
After the lesson, consider: (1) Did the LED discharge provide sufficient visual evidence, or did some groups have construction failures that disrupted the investigation — if so, how will you support those groups to rebuild and retest in the next lesson opening? (2) Did learners make the connection between the dielectric role of the plastic bag and the air gap acting as a dielectric in the storm cloud, or did the word dielectric remain abstract? (3) Were the practice problems pitched at the right level — did learners who struggled with Coulomb's Law calculations in Lesson 3 also struggle here, and what scaffolding might help? (4) Did the DQB sticky notes show genuine conceptual growth, or were learners copying language from the board without understanding? (5) How many learners independently made the transfer that the storm cloud is a natural capacitor with very large plate area — this transfer indicates readiness for the energy calculation in Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We constructed a parallel-plate capacitor from aluminium foil and a plastic bag, charged it from a 9 V battery, and discharged it through an LED. We observed that greater plate area produced a brighter or longer LED flash, and that greater plate separation produced a dimmer or shorter flash. We also observed that the LED lit up after the battery was disconnected, proving energy had been stored in the capacitor.
What did I learn?	A capacitor stores electric charge on two conducting plates separated by an insulating dielectric material. Capacitance (C) is defined as the charge stored per unit potential difference: $C = Q/V$, measured in farads (F). Capacitance increases with larger plate area and decreases with greater plate separation. The dielectric material also affects how much charge can be stored. The storm cloud above Ol Kalou acted as a natural parallel-plate capacitor, with the cloud base and the iron roof as its two plates and the air as its dielectric.
How does this explain the phenomenon?	Before the lightning strike at Ol Kalou, charge separation between the storm cloud and the iron roof created a situation identical to a charged capacitor — two conductors at different potentials separated by a dielectric (air). As the potential difference grew, so did the stored charge ($Q = CV$). When the electric field in the air gap became strong enough to ionise the air, the capacitor discharged catastrophically as a lightning bolt. The energy stored in this natural capacitor — which we will calculate in Lesson 8 — was large enough to melt iron sheet and generate the static that crackled in the nearby radio.

LESSON 7 (80 minutes): How Does Charge Store Energy — and How Can That Energy Be Released or Harnessed?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how separated electric charges store potential energy, how that energy is released (as in a lightning strike), and how capacitors harness this ability in Kenyan technology.
Knowledge	Students will know that: (1) work is done to separate charges, storing electric potential energy; (2) electric potential (voltage) is the energy per unit charge at a point in a field; (3) a capacitor stores charge and energy between two conducting plates separated by an insulator; (4) the energy stored depends on capacitance and voltage ($E = ½CV^2$); (5) lightning is the catastrophic release of stored electrostatic energy built up between cloud and ground.
Skills	Students will be able to: (1) calculate electric potential energy and voltage in simple scenarios; (2) construct and test a simple parallel-plate capacitor model; (3) represent energy storage graphically on a Q-V graph; (4) connect the concept of stored electrostatic energy to the Ol Kalou lightning event and to practical devices such as camera flash units and surge protectors.
Attitudes	Students will appreciate the value of curiosity and careful reasoning in linking abstract energy concepts to real-world hazards and protective technologies relevant to Kenyan communities.
Key Inquiry Question	How do charged objects exert forces on one another and on neutral objects at a distance?
Purpose in Storyline	Lessons 1–6 established how charges build up, how they exert forces, and how electric fields map those forces through space. Lesson 7 now asks: what happens to all that work done pushing charges apart? Where does the energy 'go', and how is it held — and eventually unleashed — as in the Ol Kalou lightning strike? This lesson is the energy-accounting chapter of the storyline: students discover that the invisible charge separation in a storm cloud is simultaneously an invisible energy reservoir, and that capacitors are engineered versions of exactly this natural phenomenon.
Safety Notes	All capacitor demonstrations must use capacitors rated at safe low voltages (≤12 V DC from a battery pack). No mains-voltage capacitor work. Van de Graaff generator sparks are safe at the distances used but students with pacemakers or hearing aids must observe from ≥1 m. Remind students never to approach a real lightning strike site or touch any conductor that may still carry residual charge. Discharge all capacitors fully before handling.

B. LESSON OVERVIEW

Students begin by predicting where the 'energy' in a lightning bolt comes from, connecting back to the Ol Kalou anchoring phenomenon. They then conduct a structured evidence-gathering sequence: (1) a teacher demonstration showing a charged Van de Graaff generator deflecting a hanging pith ball — requiring measurable work — to build intuition for electric potential energy; (2) a hands-on parallel-plate capacitor investigation using aluminium foil sheets, cardboard spacers, and a low-voltage battery pack, measuring charge stored at different voltages with a galvanometer; and (3) a data-analysis activity plotting Q vs V and computing stored energy from the area under the graph. The Explain phase formalises electric potential ($V = W/q$), capacitance ($C = Q/V$), and stored energy ($E = ½CV^2$), all anchored to the cloud-to-ground charge separation at Ol Kalou. Students update the class Driving Question Board and revise their cumulative field-and-energy models, adding energy storage notation to the lightning-conductor diagram they have been building across the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Capacitors and capacitance Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown a projected still image: a split-screen of (left) the scorch mark on the Ol Kalou iron-sheet roof and (right) a fully charged capacitor in a camera flash unit just before it fires. The teacher poses two written prediction prompts that students answer individually in their science notebooks for 3 minutes before any discussion: (A) 'Where do you think the energy in the lightning bolt at Ol Kalou came from — write a sentence and draw a simple sketch.' (B) 'The camera flash fires in 1/1000 of a second but the battery charges it over 3 seconds. What does that tell you about where the energy was sitting just before the flash?' Students then share predictions with a shoulder partner (Think-Pair) for 2 minutes. The teacher cold-calls 4 pairs (selecting one confident and one hesitant voice per pair) and scribes key vocabulary that emerges — 'stored', 'built up', 'cloud', 'separated charges' — on the left column of the Driving Question Board under the heading 'What we THINK we know (Lesson 7)'. No predictions are corrected at this stage.	Display the split-screen image silently for 30 seconds before speaking — allow visual processing. Say: 'Look carefully at both images. Do not talk yet. What do they have in common?' [WAIT 30 s]. Distribute prediction slips (or direct students to page 7 of their notebooks). Say: 'Write your prediction for prompt A, then prompt B. You have 3 minutes — full sentences, a sketch counts as evidence.' [WAIT 3 min]. Circulate and scan notebooks; place a small tick on two notebooks with particularly generative misconceptions to revisit in the Explain phase. After pair-share, cold-call exactly 4 pairs: 'Akinyi and Mwenda — what did you agree on?' / 'Chebet and Ochieng — where did you disagree?' / 'Fatuma and Kipchoge — what word kept coming up in your conversation?' / 'Muthoni and Baraka — what question did your prediction leave unanswered?' Scribe verbatim on the DQB. Do NOT evaluate — say: 'We are collecting ideas, not answers yet.'	Think-Pair-Write-Share (prediction anchoring): Students externalise prior knowledge in writing before social influence shapes their thinking. Scribing on the DQB makes predictions public and accountable — students can revisit them at the end of the lesson to measure their own conceptual shift. This strategy directly addresses the common misconception that lightning energy 'comes from the sky' rather than from work done separating charges over time.	Observable indicator: Every student has a written prediction and a sketch in their notebook before pair-share begins. Teacher scans for: (a) students who draw the cloud and ground as a 'battery-like' separated system (evidence of prior field understanding from Lesson 6); (b) students who write 'energy from friction/rubbing' — a productive partial model to build on; (c) students who leave prompts blank — these students receive a quiet prompt: 'Think back to our Ol Kalou storm — what was building up in the cloud?'
Observe Phase 📹 **VIDEO:** Capacitors and capacitance Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge	OBSERVE ACTIVITY A — Van de Graaff Work Demonstration (15 min): Teacher charges the Van de Graaff generator (or a large Leyden jar alternative if unavailable) and suspends a lightweight pith ball on a 30 cm thread from a retort stand 15 cm from the dome. As the generator charges, students observe the ball deflecting progressively further — a visible proxy for increasing	ACTIVITY A: Before switching on the generator say: 'Watch the pith ball — your job is to be a silent observer for 60 seconds. Write one word every 15 seconds describing what you see.' [WAIT 60 s while generator charges]. After discharge: 'What happened to all that deflection? Where did the energy go? Turn and tell your neighbour in one sentence.' [WAIT 15 s]. Cold-call 2 students.	Progressive Disclosure with Concrete-Representational-Abstract (CRA) Progression: Students first see a physical, visible proxy for energy storage (pith ball deflection — Concrete), then build and measure a real capacitor (still Concrete but quantitative), then extract meaning from a Q-V graph (Representational), and finally connect the graph area to the	Observable indicators: (1) Every group's capacitor produces a measurable LED flash or galvanometer deflection — if not, teacher checks crocodile clip connections and foil overlap before moving on. (2) Students correctly label 'stored state' on their Activity A sketches — teacher checks 3 random notebooks during Activity B set-up. (3) Q-V graphs pass through the origin with a positive

and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	potential. Teacher then touches the dome briefly with a grounded wire: ball snaps back to vertical. Students sketch the sequence (before/during/after charge) in their notebooks, labelling 'work input', 'stored state', and 'energy release'. OBSERVE ACTIVITY B — Parallel-Plate Capacitor Investigation (25 min): In groups of 4, students construct a parallel-plate capacitor from two 15 cm × 15 cm squares of aluminium cooking foil pressed against opposite faces of a 2 mm cardboard sheet (a common Kenyan school material). Connecting leads are attached with crocodile clips. Students connect the capacitor to a 1.5 V AA battery for 10 seconds, then disconnect and briefly connect a small LED or galvanometer — observing a flash/deflection indicating stored charge. They repeat at 3 V (two batteries in series) and 4.5 V (three batteries). For each voltage, they record: supply voltage (V), estimated deflection on galvanometer (as a relative measure of Q), and whether the LED flashes brighter. Groups record all data in a shared class table on the board. OBSERVE ACTIVITY C — Data Analysis — Q vs V Graph (10 min): Each student plots a Q (relative, in galvanometer divisions) vs V graph from the class data table. They draw the best-fit straight line through the origin and calculate the gradient ($C = Q/V$). They shade the triangular area under the line and annotate it: 'Area = stored energy = ½QV'.	ACTIVITY B: Distribute materials kits (pre-packed in labelled envelopes). Say: 'Your task is to store charge in your homemade capacitor. Before you connect anything, predict — will more voltage store more or less charge? Write it down.' [WAIT 10 s for written prediction]. As groups work, circulate with the following targeted questions — ask each group a different one: 'How do you know charge is stored and not just passing through?' / 'What is the cardboard doing — why not just touch the two foil sheets together?' / 'If you made the cardboard thicker, what would change?' / 'How is your foil-and-cardboard device similar to the cloud and ground at Ol Kalou?' After all groups record data: 'Pause all work. Let's build our class data table.' Cold-call one reporter per group to read data aloud. ACTIVITY C: Say: 'The area under a force-distance graph gives you work done. The area under a Q-V graph gives you — discuss with your partner.' [WAIT 15 s]. Cold-call 3 students before confirming.	formula $E = \frac{1}{2}CV^2$ (Abstract). This three-rung ladder prevents the common failure mode of jumping straight to formulas without physical grounding. The Ol Kalou connection is explicitly prompted at each rung so the phenomenon remains the engine of inquiry rather than a decorative hook.	gradient — a graph with a negative gradient or non-linear curve signals a data recording error; teacher pauses that group and asks: 'At 0 volts, how much charge would you expect to store?' to self-correct. (4) Exit check: teacher asks the whole class 'Thumbs up if your graph is a straight line through the origin' — students with thumbs sideways or down receive peer support before the Explain phase begins.
Explain Phase 📹 **VIDEO:**	The teacher leads a structured whole-class explanation sequence	Project the class Q-V graph prominently throughout. For the	Concept-Mapping via Sequential Anchoring: Each new concept	Observable indicators: (1) Worked example: all students who were

Capacitors and capacitance Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(15 min) using the class Q-V graph projected on the board as the central artefact. The sequence has four beats: BEAT 1 — Electric Potential Energy: Teacher formalises that work done moving a charge against an electric force is stored as electric potential energy (EPE), using the analogy of lifting a sack of maize to a raised store platform in a Rift Valley granary — height stores gravitational PE; separation stores EPE. BEAT 2 — Electric Potential (Voltage): Teacher introduces V = W/q (potential = work per unit charge), worked example: 'If 6 J of work is done moving a 2 C charge from the ground to the base of an Ol Kalou storm cloud, what is the potential difference?' Students solve in notebooks [WAIT 60 s], then cold-call. BEAT 3 — Capacitance: Teacher defines C = Q/V, students calculate the capacitance of their foil capacitor from their own graph gradient — making the abstract formula personally meaningful. BEAT 4 — Stored Energy: Teacher derives E = ½CV² from the Q-V graph area (the triangle = ½ × base × height = ½QV = ½CV²). Students solve a context problem: 'A lightning strike transfers 5 C of charge through a potential difference of 300 MV. Calculate the energy released — how many 100 W bulbs could this power for 1 hour?' Students work individually [WAIT 3 min], then peer-mark. Teacher closes by returning to the Ol Kalou image: 'The energy that scorch-marked that iron roof had been stored — invisibly, silently — in the separated charges between cloud and ground for up to 30 minutes. Your foil capacitor did exactly the	maize-store analogy say: 'Imagine you are a farmer in Nakuru lifting a 50 kg sack of maize to a platform 2 m high. You've done work. Where has that energy gone? [WAIT 10 s, cold-call 2 students]. Now replace the sack with a charge and the platform with a higher potential — same idea.' For the worked example, say: 'Show me your working in your notebook — I want to see W = qV rearranged before I see the number.' Circulate and mark a tick or a dot (dot = needs revision) on 6 books. For the lightning calculation, scaffold with: 'First write down what you know. Then write the equation. Then substitute. You have 3 minutes.' [WAIT 3 min]. After peer-marking: 'Raise your hand if your partner's answer was within 10% of yours.' Investigate outliers. Close by cold-calling one student to re-explain the Ol Kalou connection in their own words: 'Tell me the Ol Kalou story using the words potential difference, capacitance, and stored energy — go.'	(EPE → V → C → E) is explicitly anchored to the previous one and to the Q-V graph artefact students generated themselves. This 'chaining' approach ensures students build a connected schema rather than isolated formulae. The Ol Kalou closing narrative activates the anchoring phenomenon as a retrieval cue that binds all four concepts together in memory. The peer-marking step adds low-stakes accountability and surfaces calculation errors before the summative assessment.	dotted (need revision) during circulation should self-correct after the cold-call. Teacher re-checks those 6 books before the DQB phase. (2) Lightning calculation: correct answer is approximately 1.5×10^9 J; a 100 W bulb for 1 hour uses 3.6×10^5 J — so approximately 4,167 bulbs. Students who obtain an answer in this order of magnitude demonstrate correct substitution. (3) Final cold-call re-narration: the nominated student must use all three target terms correctly in context — if they use 'voltage' correctly but misplace 'capacitance', teacher asks the class: 'Who can help Amina refine that sentence?' — not correcting directly.

	same thing, just at a trillion times smaller scale.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Capacitors and capacitance](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electric Charge and Electric Force (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Students individually write one new sticky note (or index card) for each of the three DQB columns: (1) WHAT WE NOW KNOW — one sentence using a formula or key term from today; (2) WHAT WE'RE STILL WONDERING — one genuine unresolved question; (3) HOW THIS CONNECTS TO OL KALOU — one sentence linking today's energy concept directly to the anchoring phenomenon. Students post their cards on the class DQB (a section of the classroom wall or a large sheet of sugar paper). The teacher facilitates a 5-minute gallery walk where students read three other groups' 'Still Wondering' cards and place a small star sticker on any question they also have. The most-starred questions are read aloud by the teacher and designated as 'Questions to carry into Lessons 8 and 9', which will address electrostatic applications — including lightning conductors and capacitors in Kenyan infrastructure.	Distribute three sticky notes per student (or direct to notebook if materials are limited). Say: 'You have 4 minutes — no talking, this is individual thinking. I want to see a formula or key term on your KNOW card — not just a vague sentence.' [WAIT 4 min]. For the gallery walk: 'Move silently. Read — don't talk. Star the questions that are also yours.' [WAIT 5 min]. Collect the highest-starred questions and read aloud with genuine curiosity: 'Six of you are asking how a capacitor in a mobile phone charger works — that is exactly where we are going in Lesson 8. Three of you are asking whether the Ol Kalou lightning conductor acts like a capacitor — brilliant question. We'll test that idea in Lesson 9.' This move validates student-generated inquiry and signals that their questions drive the next lessons.	Three-Column DQB Protocol with Star Voting: The three-column structure forces students to distinguish between declarative knowledge, productive uncertainty, and phenomenological connection — three cognitively distinct acts. Star voting transforms individual questions into communal intellectual priorities, giving students genuine agency in the direction of the unit. This is particularly important in a competency-based curriculum where learner agency is a core KICD principle.	Observable indicators: (1) Every student posts at least one card in each column — teacher scans the board before the gallery walk and prompts any student with a missing column: 'You've got KNOW and OL KALOU — what are you still wondering about stored energy?' (2) KNOW cards that contain only vague language ('energy is stored') without a formula or specific term indicate surface-level understanding — teacher photographs these for follow-up in Lesson 8 warm-up. (3) STILL WONDERING cards that ask questions beyond the current lesson ('Can you store charge in a fruit?' / 'Does a thunderstorm capacitor have a capacitance we can calculate?') are flagged as evidence of high-order extension thinking.
Model Building Phase 📹 **VIDEO:** [Capacitors and capacitance](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electric Charge and Electric Force (Flexbook)](#)	Students retrieve their cumulative unit model — a diagram of the Ol Kalou scene that they have been building since Lesson 1 (showing charge build-up, field lines, Coulomb forces, and field strength gradients from previous lessons). In today's revision, each student adds an energy layer to the model using a coloured pen (ideally red, distinct from previous additions): (1) Label the cloud-to-ground gap as a 'natural capacitor' with approximate plate separation and estimated voltage (300 MV); (2)	Distribute coloured pens (or instruct students to use red ink). Say: 'Open your unit model to the Ol Kalou diagram. You have 8 minutes to add today's energy layer. Use red so we can see exactly what Lesson 7 contributed.' [WAIT 8 min]. Circulate with targeted questions: 'Where is the energy stored in your model — point to it.' / 'Your field lines from Lesson 6 are already there — how does stored energy relate to the field between cloud and ground?' / 'What does your model predict	Cumulative Model Revision with Colour-Coded Layering: Using a new colour for each lesson's additions makes the growth of understanding literally visible — students and teachers can see at a glance which concepts were added when and how they build on each other. This is a form of 'knowledge tracking' that aligns with the KICD CBE emphasis on visible learning progression. The non-attributive error analysis ('this model raises an interesting question') maintains psychological	Observable indicators: (1) All four model additions (natural capacitor label, energy reservoir symbol, heat/light/sound discharge arrow, lightning conductor pathway) are present — teacher uses a 4-point quick-check rubric (1 point per element) during circulation and notes any student scoring ≤2 for targeted support in Lesson 8. (2) The energy reservoir triangle (shaded area echoing Q-V graph) correctly appears in the cloud-ground gap, not in the cloud alone or the ground alone —

Source: CK-12 🔍 Search ARES for similar readings	Draw a 'stored energy reservoir' symbol (a shaded triangle, echoing the Q-V graph area) in the cloud-ground gap and annotate it with $E = ½CV^2$; (3) Add an arrow from the reservoir to the scorch mark on the iron roof labelled 'energy released as heat, light, and sound during discharge'; (4) Add a second energy pathway arrow to the lightning conductor on the district offices, labelled 'controlled discharge — energy safely conducted to ground'. Students then do a Model Gallery Walk in pairs, comparing their models and using the sentence stem: 'I added ______ because ______ and this connects to Ol Kalou because ______.' The teacher circulates and selects two models to display under a visualiser or photograph for the class — one that correctly shows both discharge pathways and one that shows a common error (e.g., energy arrow pointing the wrong way), using the error model as a non-attributive teaching moment: 'This model raises an interesting question for us — look at where this arrow is pointing. What would have to be true for that to be correct?'	would happen to the stored energy if the cloud discharged slowly rather than in one bolt?' After pair comparison: 'I'm going to show two models under the visualiser. Your job is not to judge — your job is to ask one question of each model.' Display models. Take questions from the floor (cold-call 3 students per model). Close: 'Your model is now carrying six lessons of evidence. In Lesson 8, we will ask — how do engineers in Kenya design systems that deliberately use or prevent this energy storage?'	safety while using mistakes as productive data — a key practice in competency-based formative assessment.	misplacement indicates a misconception that energy resides in one plate rather than in the field between them. (3) During pair comparison, the sentence stem is completed with a specific formula reference ('I added $E = ½CV^2$ because the cloud-ground system stores energy like a capacitor') — vague completions ('I added energy because it stores things') are noted for follow-up.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. ENERGY INTUITION: During the Predict phase, how many students spontaneously connected 'stored energy' to the work done separating charges (as opposed to seeing energy as simply 'in' the lightning bolt)? What does this tell you about the effectiveness of the field-line and Coulomb force scaffolding from Lessons 4–6?

2. CAPACITOR INVESTIGATION: Did all groups successfully observe a galvanometer deflection or LED flash from their homemade capacitor? If some groups did not, what was the most common failure mode (poor foil-cardboard contact, inadequate voltage, loose crocodile clips)? How could the materials kit be improved for Lesson 8?

3. FORMULA GROUNDING: In the Explain phase, did students approach $E = ½CV^2$ as a meaningful summary of the Q-V graph area, or as a new formula to memorise? Students who calculated C from their own graph gradient are more likely to own the formula — note which students did this successfully and which relied on copying from the board.

4. DQB QUALITY: Review the 'Still Wondering' sticky notes. Are students asking questions that point toward applications (lightning conductors, capacitors in phones,

surge protectors in Nairobi hospitals) — which is the territory of Lessons 8 and 9 — or are they revisiting unresolved questions from earlier lessons (charge induction, field line direction)? The latter signals that foundational gaps remain and should be addressed in the Lesson 8 warm-up before advancing.

5. MODEL COMPLETENESS: How many students correctly drew both discharge pathways (uncontrolled to iron roof AND controlled via lightning conductor to ground)? Students who only drew one pathway may not yet have fully integrated the comparative structure of the Ol Kalou phenomenon. Consider pairing these students with students who drew both pathways during Lesson 8 model sharing.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed: (1) a pith ball deflecting progressively as a Van de Graaff generator charged — then snapping back to vertical upon discharge; (2) a homemade aluminium-foil parallel-plate capacitor storing charge at three different voltages (1.5 V, 3 V, 4.5 V) and releasing it to flash an LED or deflect a galvanometer; (3) a class Q-V graph forming a straight line through the origin, with a shaded triangular area representing stored energy.
What did I learn?	Students learned that: (1) work done separating charges is stored as electric potential energy; (2) electric potential (voltage) equals work done per unit charge ($V = W/q$); (3) capacitance ($C = Q/V$) measures how much charge a device stores per volt; (4) energy stored in a capacitor equals the area under the Q-V graph ($E = \frac{1}{2}CV^2$); (5) the cloud-to-ground gap in the Ol Kalou storm acted as a natural parallel-plate capacitor storing up to 1.5×10^9 J before catastrophic discharge.
How does this explain the phenomenon?	Students explained that: (1) the energy scorch-marking the iron-sheet roof at Ol Kalou had been stored invisibly in the electric field between the charged storm cloud and the ground — exactly as energy is stored in the field between the foil plates of their homemade capacitor; (2) the lightning conductor on the district offices provided a controlled discharge pathway, safely routing stored energy to ground rather than through the iron roof; (3) engineers exploit the same charge-storage principle in camera flash units, mobile phone chargers, and surge protectors — devices that charge slowly and release energy rapidly or safely absorb excess energy.

LESSON 8 (40 minutes): How Much Energy Can a Capacitor Store? (Energy Stored in a Capacitor)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To quantify the electrical energy stored by a capacitor and connect that quantity to both engineered technologies and the catastrophic energy released by the Ol Kalou lightning bolt
Knowledge	Learners will be able to: (1) state and write the formula E = half CV squared; (2) derive E = half CV squared from the area under a Q-V graph; (3) calculate energy stored in single and combined capacitor configurations; (4) estimate and compare the energy in a lightning discharge to that stored by engineered capacitors
Skills	Learners will be able to: (1) analyse Q-V graphs to extract stored energy as a geometric area; (2) manipulate and apply E = half CV squared in multi-step calculations; (3) compare energy values across vastly different scales using scientific notation; (4) evaluate how capacitor energy storage is engineered for specific Kenyan applications
Attitudes	Learners will appreciate the power of mathematical models to explain natural phenomena that appear mysterious, and will value the role of physics in designing technologies that protect communities and clean the environment
Key Inquiry Question	How do charged objects exert forces on one another and on neutral objects at a distance? (extended to: how does the energy stored in separated charges get released and how do engineers control that release?)
Purpose in Storyline	This lesson is the quantitative climax of the unit. Learners now put a precise number on the energy stored in the Ol Kalou storm cloud and compare it to engineered capacitors, answering why the lightning strike was so destructive while also showing how the same physics, at controlled scales, powers defibrillators, camera flashes, and electrostatic precipitators at Athi River. The Driving Question Board gains its first fully quantitative evidence card.
Safety Notes	Ensure the foil capacitor charged in Lesson 6 is discharged through a 1 kilo-ohm resistor before handling. Do not use mains electricity. The 9 V battery circuit is safe for Grade 10. Remind learners never to touch both terminals of a large charged capacitor simultaneously. Reinforce that the lesson numbers from lightning strikes justify all the safety engineering studied in the unit.

B. LESSON OVERVIEW
Learners open with a phenomenon reconnect: the teacher replays the 30-second Ol Kalou video clip and asks one question — how much energy did that bolt carry? In the Predict phase, groups use their Q-V graph knowledge (Lesson 5 and 6) to predict the formula for stored energy by reasoning about the triangular area under a Q-V graph. In the Observe phase, they analyse discharge data from the foil capacitor charged to 9 V and plot a Q-V graph, measuring the triangular area to extract stored energy, then verify with E = half CV squared. In the Explain phase, the class derives E = half CV squared algebraically from W = QV and Q = CV, solves graded Kenya-context problems including an Athi River electrostatic precipitator calculation, and estimates the Ol Kalou lightning bolt energy from given current and duration data. The DQB Creation phase adds a quantitative evidence card. The Model Building phase updates group particle-field-energy models to show energy as stored work done against the electric field between capacitor plates.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Volume flow rate and equation of continuity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(8 minutes) The teacher replays the 30-second Ol Kalou video clip. Each learner writes one number prediction on a sticky note: how many joules do you think the lightning bolt released? Estimates are posted on the DQB without judgment. Groups then receive a pre-drawn Q-V graph (Q on y-axis in microcoulombs, V on x-axis in volts, straight line from origin to point Q-max, V-max) representing a charging capacitor. The teacher asks: if work done equals Q times V for a tiny bit of charge, what does the area under this whole line represent? Groups discuss for 3 minutes and write their area-formula prediction (most will write a rectangle; some will correctly identify the triangle). A selected group shares, prompting the class to decide: rectangle or triangle, and why?	Replay the video clip on a laptop or projector with volume on. After the clip, say: Before we do any mathematics today, I want every person to write a number — your gut estimate of the energy in that bolt. Do not consult your neighbour yet. Write it down. [WAIT 60 seconds in silence.] Post your sticky note on the DQB energy column. Now look at this Q-V graph on your sheet. Talk with your group: what is the area of the region under the line, and what unit would that area have? [WAIT 3 minutes for genuine group reasoning before calling on groups.] When a group says rectangle, ask: Is the capacitor fully charged the whole time it is charging, or does charge build up gradually from zero? [WAIT 30 seconds.] Guide them to see the triangular region. Do not confirm or deny the formula yet — say: Keep that triangle idea. The observation phase will test it.	Sticky-note energy estimation activates prior intuition and creates a public record for later revision. The Q-V area task bridges Lesson 5 graphical work to today's formula, using geometric reasoning before algebra. The deliberate silence after the video allows every learner to form an independent prior belief, which increases cognitive engagement during observation.	Teacher scans sticky-note energy estimates to gauge the range of prior knowledge (expected range: 1 J to 1 000 000 J, showing wide uncertainty). Teacher listens to group discussions of Q-V area and notes whether learners distinguish rectangle from triangle, which reveals whether the concept of incremental work (W = QV only when Q is constant) has been consolidated from Lesson 5.
Observe Phase 📹 **VIDEO:** Volume flow rate and equation of continuity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(10 minutes) Each group receives: (1) a data table of eight (Q, V) pairs recorded during the discharge of the foil capacitor from Lesson 6 (charged to 9 V, C = 470 microfarads), (2) a pre-printed graph grid, and (3) a ruler. Task 1 — learners plot the Q-V points, draw the best-fit straight line through the origin, and shade the triangular area. Task 2 — they calculate the area of the triangle using Area = half times base times height, recording their answer in microjoules. Task 3 — they independently calculate E using E = half times C times V squared with the given values (expected: E = half times 470 times 10 to the power negative 6 times 9 squared	Distribute printed data sheets. Say: You have two tasks running in parallel — a graphical method and a formula. Work through both and compare. I am not going to tell you whether they agree until you have finished both. [WAIT 5 minutes while circulating.] If a group calculates the rectangle area instead of the triangle, ask: At the very start of charging, how much charge is on the plates? So does the full voltage V-max exist from the beginning? [WAIT 20 seconds.] If a group finishes early, prompt: Can you express the triangle area in terms of Q-max and V-max using algebra? What letter does Q divided by V equal? This scaffolds the algebraic	The dual-method convergence (geometric area equals formula result) is the central sense-making move of the observation phase. When two independent methods yield the same answer, learners experience mathematical coherence as evidence. The foil capacitor from Lesson 6 provides continuity of the physical object, reinforcing that the abstract formula belongs to a real device they built.	Teacher checks whether both methods give approximately 19 mJ (tolerance of plus or minus 2 mJ is acceptable given graph-reading error). If groups get discrepant values, this diagnoses either: graph plotted incorrectly (check axis labels and scale), incorrect triangle area formula (rectangle error), or unit conversion error (microfarads not converted). Teacher notes these for targeted support during the Explain phase.

	= approximately 19 mJ) and compare the two answers. Task 4 — they record whether the two methods agree and write one sentence explaining what the triangle represents physically (stored work done by the battery in separating charges onto the plates).	derivation in the next phase. Bring the class together after 8 minutes and ask two groups to share their energy value from each method: the graphical method and the formula method.		
Explain Phase 📹 **VIDEO:** Volume flow rate and equation of continuity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(12 minutes) Part A — Algebraic Derivation (4 minutes): The teacher guides the class through a step-by-step derivation on the board. Step 1: the energy stored equals the area of the triangle under the Q-V graph, so E = half times Q times V. Step 2: since Q = CV, substitute to get E = half times CV times V = half CV squared. Step 3: alternatively, substitute V = Q divided by C to get E = Q squared divided by 2C. Learners copy the three equivalent forms and annotate each variable with units and physical meaning. Part B — Graded Problem Set (8 minutes): Groups solve three problems. Problem 1 (basic): A 100 microfarad capacitor is charged to 12 V (as in a small Kenyan solar-lantern circuit). Calculate the energy stored. Problem 2 (intermediate): A camera flash unit uses a 2 200 microfarad capacitor charged to 300 V. Calculate the energy stored and compare it to your Lesson 6 foil capacitor — which stores more energy and by what factor? Problem 3 (applied): An electrostatic precipitator at the Athi River cement factory uses a bank of capacitors with total capacitance 0.05 F charged to 50 000 V to create the electric field that attracts dust particles. Calculate the energy stored. The teacher then	For Part A, write each substitution step slowly and say: Watch what I am substituting and why. [WAIT 10 seconds after each step before writing the next.] Do not rush this derivation — the algebraic manipulation is new for many learners. After writing E = half CV squared, ask: Which variable has the strongest effect on stored energy — C or V? Why? [WAIT 30 seconds.] Expect learners to note the squared term on V. For Problem 3, after groups calculate the Athi River answer (62 500 J), ask: Compare that to the lightning bolt energy. Is the precipitator storing more or less? What does that tell you about why lightning is so dangerous? [WAIT 45 seconds.] For the bolt energy estimate, provide the numbers on the board and let groups calculate without scaffolding first. After 2 minutes, ask a group to share their method before revealing the answer.	The three-form derivation (E = half QV, E = half CV squared, E = Q squared over 2C) ensures learners understand the formula as a relationship, not a single equation to memorise. The graded problems move from familiar (solar lantern) to industrial (Athi River) to natural (lightning), creating a coherent scale ladder. Comparing student sticky-note predictions to the calculated bolt energy creates a visible revision moment that celebrates scientific estimation as a skill.	Teacher collects Problem 1 answers (expected: 7.2 times 10 to the power negative 3 J = 7.2 mJ) to check unit conversion fluency. For Problem 3, teacher checks whether learners correctly handle 0.05 F (not microfarads) and 50 000 V squared. Errors in Problem 3 typically reveal either unit-conversion failure or calculator-entry errors with large numbers — teacher addresses these by asking the group to estimate the answer in powers of ten first before calculating precisely.

	reveals the Ol Kalou lightning bolt data: peak current 30 000 A, duration 0.2 milliseconds, effective resistance of air channel approximately 10 ohms. Learners use P = I squared R and E = Pt to estimate bolt energy (approximately 1.8 times 10 to the power 6 J) and compare to their sticky-note predictions.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Volume flow rate and equation of continuity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(5 minutes) Each group receives one large index card. They write a DQB evidence card with four elements: (1) the formula E = half CV squared with a labelled diagram of the triangle area on a Q-V graph; (2) the energy stored in the Ol Kalou storm-cloud capacitor (from Lesson 5: E = approximately 1.5 times 10 to the power 9 J); (3) the energy stored in the Athi River precipitator bank (approximately 6.25 times 10 to the power 4 J); (4) one sentence connecting energy storage to the Driving Question. Groups post their cards in the Energy column of the DQB. The teacher invites one group to read their connecting sentence aloud and asks the class: Has this formula helped us explain why the Ol Kalou bolt was so destructive and why the lightning conductor needed to redirect it safely? What part of the Driving Question can we now answer more completely?	Circulate as groups write their cards and check that all four elements are present. Prompt groups who only write the formula: What numbers can you put into that formula to make it real? After cards are posted, read two contrasting cards aloud (one strong, one partial) and ask the class: Which card gives us more evidence toward the Driving Question, and what is missing from the other? [WAIT 30 seconds.] Do not single out learners by name when comparing cards — make it a feature-focused discussion. End this phase by pointing to the DQB: Look at how far this board has come since Lesson 1. We now have a number for the energy that struck Ol Kalou market that day.	The four-element card structure ensures learners integrate formula, data, context, and explanation — not just a symbolic result. Displaying cards publicly in the Energy column makes the unit narrative visible and cumulative. Reading the connecting sentence aloud develops scientific communication and positions every lesson as building toward the Driving Question answer.	Teacher checks each card before posting. Cards that only contain the formula without numbers or context reveal that the learner has not yet connected the mathematics to the phenomenon — these learners are identified for targeted questioning during Model Building. Cards with incorrect energy values for the storm cloud (not consistent with Lesson 5 value) reveal that lesson-to-lesson knowledge transfer needs reinforcement.
Model Building Phase 📹 **VIDEO:** Volume flow rate and equation of continuity Source: Khan Academy (English - US curriculum) 🔍 Search ARES	(5 minutes) Learners open their individual particle-field-energy model journal (started in Lesson 1 and updated in every subsequent lesson). Today they add one new layer. They draw a parallel-plate capacitor (two lines representing plates) with plus charges on one plate and minus charges on the other, electric field lines between	Say: Take the last 5 minutes to update your personal model. I want to see the triangle on the Q-V graph — that triangle IS the energy. If you can draw it, you understand it. [WAIT in silence for 3 minutes while learners draw and annotate.] After 3 minutes, ask one learner to share their diagram under the document camera (or	The cumulative personal model journal is the most important sense-making artefact of the unit. Adding a new layer in every lesson makes conceptual growth visible to the learner. The triangle annotation enforces the geometric meaning of the formula rather than mere symbol manipulation. Placing both natural and engineered	Teacher does a rapid gallery walk during the 5-minute draw period, looking for: (1) field lines drawn between plates (not outside), (2) the triangular region correctly shaded on the Q-V inset, (3) energy values written in scientific notation with correct powers of ten. Learners missing the triangle or drawing field lines incorrectly

for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the plates, and a shaded triangular region on a small Q-V inset graph. They annotate: the work done separating charges is stored as energy in the electric field between the plates. The value of that stored energy equals the area of the triangle = half CV squared. They add an arrow from their capacitor diagram to a small sketch of a lightning bolt, labelling it: natural capacitor — storm cloud to ground — stores approximately 10 to the power 9 J, and another arrow to a small sketch of a precipitator labelled: engineered capacitor bank — stores approximately 10 to the power 4 J. Finally, they write one sentence under the Driving Question: I can now explain that invisible electric charges build up, store energy proportional to the square of the voltage, and that energy can be released catastrophically as at Ol Kalou or harnessed deliberately in technologies that protect communities.	hold it up). Ask the class: What does the shaded triangle tell us that the formula alone does not? [WAIT 30 seconds.] Expected answer: it shows that energy builds up gradually as the capacitor charges — the last bit of charge added at high voltage stores the most energy. Close the lesson by saying: Next lesson we will use everything in your model to answer the Driving Question fully — every group will present one Kenyan electrostatic application and we will complete the board together.	capacitors side by side in one diagram achieves the lesson goal of comparing scales and connecting to the phenomenon.	receive a verbal prompt: Where does the energy actually live — on the plates or in the space between them? This diagnostic informs whether to open Lesson 9 with a brief 3-minute energy-field review.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners successfully connect the triangular area to the formula without being told directly, or did most wait for teacher confirmation? If they waited, the Predict phase graphic needs a more provocative guiding question next time. (2) Were learners able to handle the large-number arithmetic in Problem 3 (0.05 F, 50 000 V) without unit-conversion errors? If not, a dedicated unit-conversion warm-up should open Lesson 9. (3) Did the sticky-note energy estimates shift meaningfully toward the calculated value? If estimates remained unchanged even after the calculation, learners may not have made the personal connection between their intuition and the mathematics — try asking learners to write a revision note on the back of their original sticky note. (4) Were DQB evidence cards genuinely evidential (containing numbers and a phenomenon link) or merely formulaic? If mostly formulaic, the DQB instruction needs to explicitly model the four-element structure in Lesson 9.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We explained that the energy stored in a capacitor is the work done by the battery in separating charges onto the plates against the growing electric field. Because voltage increases as charge builds up, the work done per extra unit of charge also increases, so the total energy is the triangular area under the Q-V graph rather than a simple rectangle. This is captured exactly by E = half CV

	squared.
What did I learn?	We learned that the energy stored in a capacitor equals half times C times V squared, that voltage has a squared effect on stored energy (doubling the voltage quadruples the energy), and that this formula applies equally to small engineered capacitors and to the enormous natural capacitor formed between the storm cloud and the ground above Ol Kalou.
How does this explain the phenomenon?	The energy stored in a capacitor is not simply charge times voltage — it is half that, because the voltage starts at zero and grows as charge accumulates, so the average voltage across the charging process is V/2. This is why doubling the voltage on a capacitor quadruples the energy it stores: the formula E = half CV squared has voltage squared, not linear. The same physics explains the enormous energy released in a lightning strike above Ol Kalou — the natural capacitor formed between the storm cloud and the ground charges up to millions of volts, and even though its capacitance is modest, the V squared term makes the stored energy catastrophic when it discharges.

LESSON 9 (40 minutes): How Do We Use Electrostatics to Solve Real Problems? (Applications and Unit Synthesis)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all unit learning by connecting charge behaviour, electric fields, potential difference, capacitance, and energy storage to real Kenyan technological applications, and produce a final written explanation of the anchoring phenomenon.
Knowledge	Learners will be able to: (1) describe how at least three electrostatic technologies function in terms of charge, field, and energy concepts; (2) explain how the lightning conductor at Ol Kalou District Offices uses field concentration at a pointed tip to safely discharge the storm cloud; (3) connect each application to the specific sub-strand concepts introduced across Lessons 1 to 8.
Skills	Learners will research, organise, and present one electrostatic application using scientific vocabulary; construct and annotate a final cumulative particle-and-field model; write a structured scientific explanation of the anchoring phenomenon using evidence from all nine lessons.
Attitudes	Learners appreciate that physics concepts developed in the classroom have direct, life-saving and economic value in Kenyan communities, and feel confident that they can explain complex invisible phenomena using a coherent particle-and-field model.
Key Inquiry Question	How do objects acquire electric charge and how does charge behave on different materials? How do charged objects exert forces on one another and on neutral objects at a distance?
Purpose in Storyline	This final lesson is the payoff of the entire nine-lesson arc. Learners return to every unanswered question on the Driving Question Board — the scorch mark on the iron roof, the untouched wooden stall, the crackling radio, the lightning conductor — and answer them using the full conceptual toolkit built since Lesson 1. Group presentations ensure peer teaching of applications while the summative writing task gives every learner an individual opportunity to demonstrate mastery of the complete explanatory narrative.
Safety Notes	No electrical apparatus is energised in this lesson. If printed circuit boards or capacitor samples are used as display artefacts, ensure discharged state is confirmed before handling. Standard classroom supervision applies.

B. LESSON OVERVIEW
Learners spend the first segment making predictions about how their chosen application works before researching it deeply. In the Observe phase, five groups deliver short structured presentations on their assigned Kenyan electrostatic applications, with peer questioning after each. In the Explain phase, the teacher facilitates a whole-class synthesis connecting every application back to the anchoring phenomenon and the driving question. The DQB receives its final update — every sticky-note question is answered and colour-coded as resolved. In the Model Building phase, each learner independently completes a summative written explanation of the Ol Kalou lightning event using all nine lessons of evidence, assessed against the phenomenon writing rubric.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Molecule	Duration: 6 minutes. Each group of four to five learners receives a single application card printed with	Distribute application cards and mini-whiteboards. Say: Before you research or present, I want your	Prediction boards activate prior knowledge from Lessons 1 to 8 and create cognitive tension —	Teacher scans held-up mini-whiteboards to identify: which concept vocabulary is appearing

Polarity Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Applications: Fluid Forces (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	a photograph and two sentences of context: Group A gets a photograph of the lightning conductor on Ol Kalou District Offices; Group B gets an image of the Athi River cement factory chimney with visible dust plumes and a description of electrostatic precipitators; Group C gets a photograph of a photocopier in a Kisumu government office; Group D gets an image of an electrostatic spray-painting booth at a Nairobi industrial park on Mombasa Road; Group E gets a diagram of an inkjet printer cartridge. Each group reads their card silently for one minute, then spends four minutes writing answers to two prompt questions on their mini-whiteboard: (1) Which concepts from our unit — charge, induction, field, potential, capacitance, energy — do you predict are involved in making this technology work? (2) How does your application connect to what happened at Ol Kalou Market? Groups do not share answers yet; predictions are held for comparison after presentations.	honest prediction. You have four minutes — use the concepts from Lessons 1 to 8 and think out loud on your board. There are no wrong answers at this stage; we are activating your existing model. After two minutes, circulate and offer a silent prompt to any group that is stuck by pointing at the DQB wall without speaking. At the four-minute mark say: Hold your boards up. Do not change anything yet. We will return to these predictions after every group presents. WAIT TIME: After asking the prediction questions, allow a full 60 seconds of silent individual thinking before group discussion begins.	learners are committed to a prediction before evidence arrives, making the subsequent presentations more meaningful. The instruction to connect back to Ol Kalou keeps the anchoring phenomenon active.	spontaneously (charge, field, induction, capacitance, energy); which groups are still reasoning purely descriptively without physics vocabulary; which groups have made the Ol Kalou connection. This informs which groups to probe with deeper questions during their presentation.
Observe Phase 📹 **VIDEO:** Video: Molecule Polarity Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Applications: Fluid Forces (Flexbook) Source: CK-12 🔍 Search ARES	Duration: 14 minutes. Each group has 2.5 minutes to present their application, followed by 20 seconds of peer questions. Groups use their application card, their mini-whiteboard prediction (now updated or confirmed), and any hand-drawn diagrams they prepared. Presentations follow a strict three-part structure written on the board: (1) Name and describe the application in a Kenyan context — one sentence. (2) Explain the physics — which concepts from our unit make this work? (3) Show the Ol Kalou connection — how does your application share the	Introduce the session: You each have 2.5 minutes and three things to cover — context, physics, and the Ol Kalou link. I will hold up one finger at the 2-minute mark as a warning. Time each group strictly to model scientific communication discipline. After each presentation ask the class: Does this group agree with their original prediction? What changed? For the comparison table, ask: Looking at this table, what do all five applications have in common? WAIT TIME: After asking what all applications have in common, count silently to 10 before taking	The structured three-part presentation format ensures every group uses scientific vocabulary rather than describing applications superficially. The live comparison table makes cross-application patterns visible and supports the final synthesis. The prediction-versus-presentation comparison creates productive cognitive conflict.	Teacher listens for: correct use of technical vocabulary (induction, field concentration, potential difference, earthing, capacitance); accurate Ol Kalou connections; misconceptions such as the idea that lightning conductors attract lightning (address immediately: the conductor lowers field concentration differences gradually, it does not summon bolts). Peer questions after each presentation reveal depth of audience understanding.

for similar readings	same underlying physics as the lightning event? Group A (lightning conductor): Describes how the pointed rod on the Ol Kalou District Offices building concentrates the electric field at its tip (Lesson 4), lowers the potential difference gradually rather than allowing sudden discharge (Lesson 5), and safely conducts charge to earth (Lesson 2 earthing concept). Group B (electrostatic precipitator): Describes how charged plates inside the Athi River factory chimney attract oppositely charged dust particles (Lessons 1 and 2), reducing air pollution — learners recall from Lesson 8 the energy cost calculation. Group C (photocopier): Describes the selenium drum acquiring charge by the corona wire (Lesson 2), the image creating a field pattern (Lesson 4), toner particles attracted by the field (Lessons 3 and 4). Group D (electrostatic spray painting): Describes how the spray nozzle charges droplets, the grounded car body creates an attractive field (Lessons 1, 2, 3), achieving even coating and reducing waste. Group E (inkjet printer): Describes how droplets are charged and deflected by electric fields between deflection plates (Lesson 4), controlled by potential difference (Lesson 5) to place ink precisely. After all five presentations, the teacher projects a comparison table on the board with columns: Application, Method of charging used, Role of electric field, Role of potential difference, Energy concept involved, Ol Kalou connection. Learners call out entries while the teacher fills the table in real time.	any response. Say: I am giving you thinking time — this is the most important question of the lesson.		

Explain Phase 📹 **VIDEO:** Video: Molecule Polarity Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Applications: Fluid Forces (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Duration: 8 minutes. The teacher facilitates a whole-class synthesis conversation using the completed comparison table and the DQB. The teacher reads aloud the original driving question — How do invisible electric charges build up, exert forces across empty space, and store energy, and how can we harness these abilities to protect communities and power technology in Kenya? — and asks learners to respond using evidence from the table. The teacher then walks through the Ol Kalou event one final time as a complete narrative: (1) Charge separation inside the cumulonimbus cloud above Nyandarua builds by friction between ice crystals and water droplets (Lesson 1). (2) The negatively charged cloud base induces positive charge on the iron roof by induction without contact (Lesson 2). (3) The Coulomb force between the cloud and roof intensifies as separation decreases with the descending stepped leader (Lesson 3). (4) The electric field between cloud and earth becomes extremely strong; the lightning conductor tip concentrates this field further (Lesson 4). (5) The potential difference between cloud and earth reaches tens of millions of volts — the air ionises and conducts (Lesson 5). (6) The cloud-earth system acts as a giant capacitor discharging rapidly (Lesson 6). (7) In the conductor network, the discharge path is equivalent to capacitors in parallel delivering maximum current (Lesson 7). (8) The energy released equals one half times C times V squared — enough to melt	Say: We started this unit with a market stall in Ol Kalou and a mystery. Right now I want to hear someone give me the complete story — from the ice crystals in the cloud to the scorch mark on the iron sheet — using the words charge, field, potential, capacitance, and energy. Choose a confident volunteer and let them narrate uninterrupted. Then say: What would you add or change? Open to the class. Probe with: Why did the wooden stall survive? (Wood is an insulator — no charge migration, no current path, no heating.) Why did the radio crackle before the strike? (Rapidly changing electric field induces current fluctuations in the radio antenna — electromagnetic induction preview.) Write both answers on the DQB as final resolved entries. WAIT TIME: Before selecting the volunteer narrator, say — Take 30 seconds, organise your nine steps in your head — then begin.	The nine-step narrative connects every lesson explicitly, making the storyline structure visible to learners. Writing lesson numbers in the margin of their notebooks gives learners a retrieval cue and demonstrates to them concretely that they have built a complete explanatory model over nine lessons. Answering the two lingering DQB questions (wooden stall, radio crackle) provides closure on the phenomenon.	The volunteer narrator reveals depth of integrated understanding. Teacher listens for whether the learner can distinguish between charging by induction (Lesson 2), field concentration (Lesson 4), and energy release (Lesson 8) rather than conflating them into a single vague step. The wooden-stall question assesses retention of conductor-insulator distinction from Lesson 1.

	the iron sheet and scorch the wood (Lesson 8). (9) The applications we have just presented all harness these same processes safely and productively (Lesson 9). Learners annotate their science notebooks with this nine-step narrative, numbering each step and writing the corresponding lesson number in the margin.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Molecule Polarity] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Applications: Fluid Forces (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Duration: 4 minutes. The class gathers around the DQB which has carried sticky-note questions since Lesson 1. The teacher distributes green dot stickers to every learner — one per person. Learners walk to the board and place their green dot on any one question they feel has now been fully answered by their own learning. The teacher then leads a 2-minute rapid-fire verbal check: pointing to each original question in turn and asking for a one-sentence answer from the floor. Questions addressed include: What invisible thing built up in the cloud and in the iron roof? (Separated electric charges — excess electrons in the cloud, deficit on the roof.) Why did the wooden stall survive? (Wood is an insulator — charge cannot flow through it to create a current, so no heating occurs.) How did the lightning conductor know where to send the charge? (It does not know — the pointed geometry concentrates the field and provides the lowest resistance path to earth, guiding the discharge.) Why did the radio crackle before the strike? (The rapidly changing electric field induced tiny currents in the radio antenna, heard as static.) New sticky notes are now added by	Say: Look at this board. Every question on it was placed here in Lesson 1 when we watched the Ol Kalou video and you did not yet have the tools to answer any of them. Today you have answered every single one. Place your green dot on the question that meant the most to you personally. After dotting, point to each question quickly: Who wants to give me one sentence? Select different learners from those who spoke in the Explain phase. End by saying: This board represents nine lessons of thinking. The photograph we take now is evidence of your learning journey. WAIT TIME: Before the rapid-fire round, say — Scan the board for 20 seconds and pick the question you feel most confident answering. Then we begin.	The green-dot activity is a kinaesthetic metacognitive reflection — learners physically mark their own sense of resolution. The rapid-fire verbal closure around original questions shows learners concretely that the inquiry cycle is complete. Adding new Resolved sticky notes in student voice reinforces ownership of the knowledge constructed.	The distribution of green dots reveals which questions learners found most meaningful or hardest to resolve. If a question receives no dots, the teacher makes a note for future lesson planning. The one-sentence answers during rapid-fire reveal remaining gaps — particularly around induction versus contact charging and field concentration concepts.

	learner volunteers in a Resolved column: Lightning conductors work by field concentration at a tip and earthing; Electrostatic precipitators use charged plates to remove pollution; Capacitors store energy as E equals half CV squared; The cloud-earth system is a giant natural capacitor. The DQB is photographed for the school records and the teacher congratulates the class on completing the explanatory journey.			
Model Building Phase 📹 **VIDEO:** [Video: Molecule Polarity] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Applications: Fluid Forces (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Duration: 8 minutes. Each learner independently completes the summative written explanation task in their science notebook. The task is projected on the board and read aloud: Using evidence from all nine lessons, write a scientific explanation of the Ol Kalou Market lightning event. Your explanation must: (1) describe how charge built up in the cloud and was distributed on the iron roof and on the lightning conductor; (2) use the terms electric field and potential difference to explain why the discharge occurred where it did; (3) use the energy formula E equals half CV squared to estimate whether the energy released is consistent with melting iron sheet (give a reasoned numerical estimate); (4) explain why the wooden stall was undamaged; (5) explain how the lightning conductor on the District Offices building prevented damage there; (6) connect one of the five group applications to the same underlying physics. Learners write continuously for 7 minutes. In the final minute, they self-assess their explanation against the four-point phenomenon writing rubric	Distribute the task by projecting it and reading each point clearly. Say: This is your individual evidence of the journey you have made since Lesson 1. Use every tool in your toolkit — particle model, field diagrams, equations, Kenyan contexts. You have seven full minutes of silence. Circulate without speaking for the first five minutes to signal that this is individual work. In minute six, quietly prompt any learner who has not yet attempted the quantitative point by whispering: Try estimating C and V for the cloud-earth system from Lesson 8 data. In minute seven say: You have 60 seconds — finish your final sentence and complete the self-assessment rubric. After time is called say: Share your self-assessed level with your neighbour and explain why you chose it — 30 seconds only. WAIT TIME: After reading the task aloud, allow 45 seconds of silent reading and planning time before writing begins.	The six-part structured explanation task scaffolds the full unit narrative without removing cognitive demand — learners must supply the causal links themselves. The self-assessment rubric makes assessment criteria transparent and develops metacognitive awareness. The neighbour-share in the final 30 seconds creates a low-stakes peer accountability moment without reducing the summative nature of the task.	The written explanations are collected at the end of the lesson and marked against the four-point rubric. Teachers look specifically for: correct use of charging by induction (Lesson 2) to explain the iron roof; field concentration at the conductor tip (Lesson 4); use of W equals QV or E equals half CV squared with numbers (Lessons 5 and 8); conductor-insulator distinction for the wooden stall (Lesson 1); and connection of at least one application to the underlying physics (Lesson 9). Self-assessed levels are compared against teacher-assessed levels to identify learners with significant overestimation or underestimation of their own understanding for follow-up.

	displayed on the board: Level 1 — describes the event with everyday language only; Level 2 — names relevant concepts but does not link them causally; Level 3 — links concepts causally using correct vocabulary throughout; Level 4 — uses quantitative reasoning (Coulombs Law, E equals half CV squared) and connects to a real application. Learners circle their self-assessed level and write one sentence about what they would add if they had more time.			

D. TEACHER REFLECTION
After the lesson consider: Did every group presentation use at least three distinct unit concepts, or did some groups remain descriptive? Were learners able to reconstruct the nine-step Ol Kalou narrative without heavy prompting, or did they still conflate induction with direct contact charging? Did the comparison table reveal any application that the class found hard to connect back to the phenomenon — if so, which concept link was weakest? Did the summative writing show quantitative reasoning (use of E equals half CV squared with numbers) or did most learners remain qualitative? Were there learners who self-assessed at Level 4 but whose writing did not justify that level — these learners may benefit from a one-to-one conversation about distinguishing description from causal explanation. Consider whether the DQB questions that received no green dots should be revisited at the start of the next sub-strand as a bridging activity. Note which Kenyan application generated the most student engagement as a pointer for future context choices in electrostatics teaching.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed five student group presentations on electrostatic applications operating in Kenya — lightning conductors in Nyandarua, electrostatic precipitators at Athi River, photocopiers in Kisumu, spray-painting booths on Mombasa Road, and inkjet printers — and we completed the Driving Question Board by placing green dots on resolved questions from Lesson 1.
What did I learn?	We learned that all five applications rely on the same fundamental processes — charge separation or induction, electric field concentration or direction, controlled potential difference, and energy transfer — that we used to explain the Ol Kalou lightning strike, confirming that electrostatics is a single coherent framework operating at every scale from a printer droplet to a storm cloud.
How does this explain the phenomenon?	We explained the complete Ol Kalou Market event in a nine-step narrative linking charge build-up by friction in the cloud, induction on the iron roof, Coulomb force intensification, field concentration at the conductor tip, ionisation of air at breakdown potential difference, rapid capacitor-like discharge releasing energy equal to half CV squared, selective damage to the conducting iron roof but not the insulating wooden stall, and the safe earthing pathway provided by the lightning conductor — and we connected this explanation to at least one engineered technology that harnesses the same physics productively for Kenyan communities.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.